

Metasurface-Driven Sensing in Terahertz Region: Principles, Applications, Fabrication, and Challenges

Xiaotong Li, Hyung-Taek Lee, Doohyuk Han, and Junsuk Rho*

The terahertz (THz) spectral region offers unique advantages for sensing owing to its long wavelengths and low photon energies, enabling non-destructive interactions with a wide range of materials, particularly suited for probing molecular vibrations and physical, chemical, and biological phenomena. However, conventional THz sensing methods often fail to detect ultrathin films or nanoscale molecular interactions, as their dimensions are significantly smaller than THz wavelengths. To address this challenge, THz metasurfaces, engineered structures with strong light-material interactions, provide pronounced and tunable spectral features highly sensitive to environmental perturbations, thereby bridging the scale mismatch and significantly enhancing detection sensitivity. By leveraging tailored resonances and near-field enhancement, THz metasurfaces facilitate sensing of biomolecules, ultrathin films, and even subtle changes in physical parameters such as temperature, strain, and pressure. This review presents a comprehensive overview of THz metasensors, structured around their fundamental physical mechanisms, key performance metrics, and sensing applications. In addition, it discusses diverse design strategies and fabrication approaches, including both mask-based and maskless techniques, for the realization of high-performance THz metasurfaces. With their exceptional sensitivity and versatility, THz metasensors hold significant promise as an enabling platform for ultra-sensitive sensing and are poised to play a pivotal role in next-generation sensing technologies.

vibrational or rotational modes in the THz range, making THz waves exceptionally effective for identifying molecular fingerprints. Consequently, THz sensing has garnered attention across various domains, such as biomedical diagnostics, chemical and environmental monitoring, material characterization, and security screening. However, conventional THz sensing techniques encounter considerable challenges in detecting micro- or trace-level samples. This is primarily due to the large wavelength of THz radiation compared to the nanoscale dimensions of most biomolecules, resulting in weak interaction and inadequate signal resolution that significantly hinders THz sensing detection capabilities.^[4–6]

Recently, rapid developments of metasurfaces have provided new opportunities for highly sensitive and selective detection.^[7–13] Metasurfaces,^[14–32] composed of planar microstructures (i.e., meta-atoms), have attracted considerable attention due to their exceptional capability to manipulate EM waves and exhibit various fascinating phenomena, including beam deflection,^[33–39]

hologram,^[40–46] metalens,^[47–59] spin Hall effect,^[60–68] and vortex beam generation.^[69–74] By customizing the geometry and configuration of meta-atoms, the THz metasurface-based sensing device, referred to as the THz metasensor, can substantially enhance the interaction between THz waves and trace analytes through a pronounced local field enhancement effect. Compared with traditional sensing platforms such as Quartz Crystal Microbalance (QCM), Surface Plasmon Resonance (SPR), and Enzyme-Linked Immunosorbent Assay (ELISA),

1. Introduction

Terahertz (THz) waves, typically defined in the frequency range of 0.1–10 THz, occupy the region between microwave and infrared radiation in the electromagnetic (EM) spectrum.^[1–3] Owing to their low photon energy and non-ionizing nature, THz waves are ideally suited for non-invasive, label-free, and non-destructive sensing applications. Furthermore, numerous chemical, biological, and physical substances exhibit characteristic

X. Li, H.-T. Lee, D. Han, J. Rho
 Department of Mechanical Engineering
 Pohang University of Science and Technology (POSTECH)
 Pohang 37673, Republic of Korea
 E-mail: jsrho@postech.ac.kr

H.-T. Lee, J. Rho
 POSCO-POSTECH-RIST Convergence Research Center for Flat Optics and Metaphotonics
 Pohang 37673, Republic of Korea

D. Han, J. Rho
 Quantum Information Science Program
 Graduate School of Convergence Science and Technology
 Pohang University of Science and Technology (POSTECH)
 Pohang 37673, Republic of Korea

J. Rho
 Department of Chemical Engineering
 Pohang University of Science and Technology (POSTECH)
 Pohang 37673, Republic of Korea

J. Rho
 Department of Electrical Engineering
 Pohang University of Science and Technology (POSTECH)
 Pohang 37673, Republic of Korea

The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/admt.202500886>

DOI: 10.1002/admt.202500886

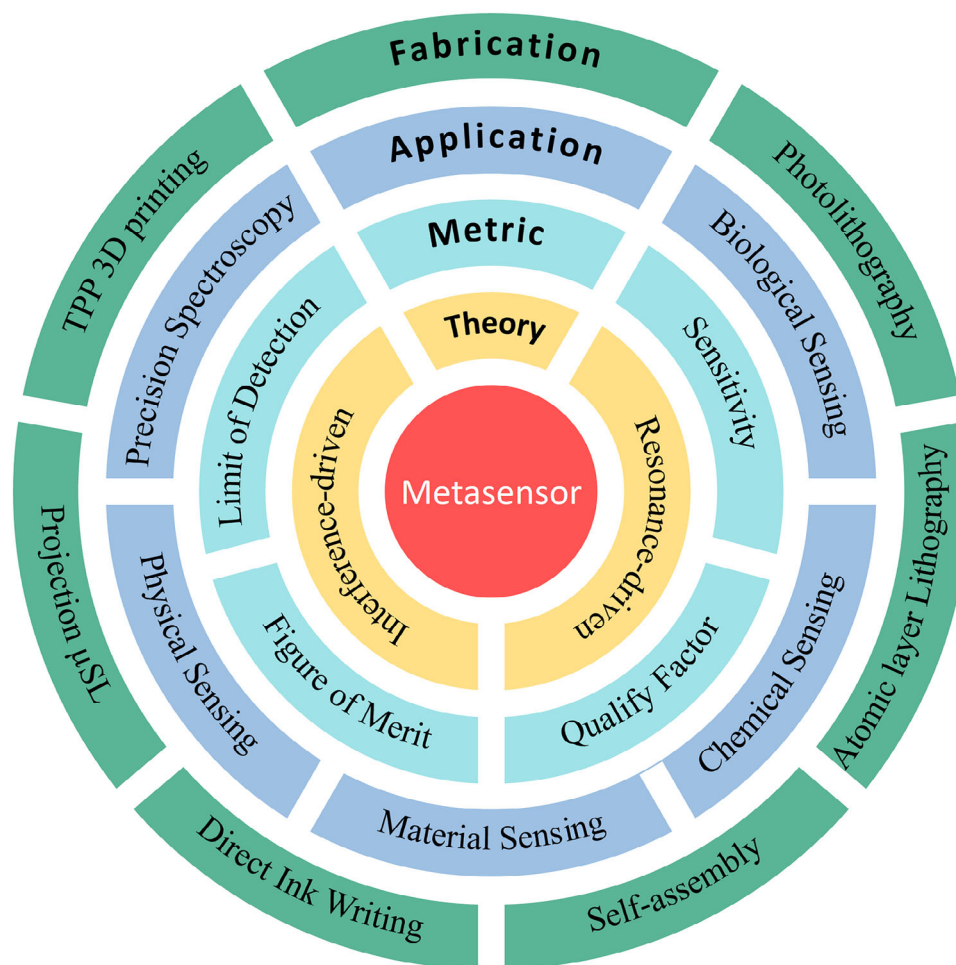


Figure 1. Schematic showing the organization of topics in this review.

THz metasensors offer competitive or superior sensitivity (S).^[75–80] ELISA commonly achieves detection limits in the 1–10 pg mL^{−1} range ($\approx 10^{-12}$ – 10^{-13} M) via enzymatic amplification, while SPR sensors typically reach limits of 1–10 ng mL^{−1} ($\approx 10^{-9}$ – 10^{-10} M) and QCM operates near 10–100 ng mL^{−1} ($\approx 10^{-9}$ M) without signal amplification, though its S can be improved to approximately 0.1 pM ($\approx 10^{-13}$ M) using nanoparticle-assisted strategies. In contrast, THz metasensors have demonstrated label-free detection capabilities extending into the picomolar (pM), femtomolar (fM), and even attomolar (aM), depending on the specific analyte and functionalization approach. While both QCM and SPR offer label-free and real-time analysis, their S is often constrained by material losses and surface noise. ELISA, although highly sensitive, involves multi-step protocols and labeling processes, which limit its throughput and integration. THz metasensors overcome these limitations by combining high-S with label-free, real-time detection in a planar and scalable architecture. The specificity of THz metasensors is primarily derived from immobilized biochemical recognition elements, such as antibodies, aptamers, or oligonucleotide probes, like the affinity-driven selectivity observed in QCM and SPR. The metasurface itself serves to amplify the sensor response upon target binding but does not intrinsically determine molecular speci-

ficity, except in specialized designs that exploit vibrational signatures or dielectric resonance matching. This distinction is critical: whereas ELISA employs a dual-antibody recognition scheme to achieve high specificity, label-free platforms, including THz metasensors, rely on single-site molecular interactions, with the metasurface enhancing detection S rather than selectivity. Moreover, THz metasensors are compatible with various integration strategies, including microfluidics, nanomaterial functionalization, and flexible substrates, which significantly expand their versatility and applicability. The planar and compact characteristics of metasensors also make them well-suited for integration into portable and on-chip platforms, facilitating the development of next-generation THz lab-on-a-chip sensing devices.

This review focuses on the recent advancements in THz metasensors and is organized as depicted in **Figure 1**. First, we begin with a concise overview of the fundamental mechanisms underlying metasensor in THz region, classifying them into resonance-driven and interference-driven. Then, we present the essential performance metrics employed to evaluate the THz metasensor, which encompass sensitivity, quality factor (Q-factor), figure of merit (FOM), and limit of detection (LoD). Subsequently, we discuss extensive applications in biological sensing, chemical sensing, material sensing, physical sensing,

and precision spectroscopy. Furthermore, we also investigate various fabrication methods for THz metasensors, including photolithography, atomic layer lithography, self-assembly, direct ink writing, projection micro-stereolithography (μ SL), and two-photon polymerization (TPP) 3D printing. Finally, we conclude this review with a discussion of the current challenges in the field and present insights into prospective advancements, with the objective of equipping researchers with a thorough comprehension of THz metasensors and their practical applications.

2. Theory

THz metasensors utilize advanced physical mechanisms to significantly enhance the interaction between target analytes and THz waves, thereby improving sensing performance. These mechanisms can be broadly classified into two categories: resonance-driven and interference-driven. Resonance-driven mechanisms include surface plasmon polaritons (SPPs), localized surface plasmons (LSPs), Mie resonances, toroidal resonances, and perfect absorbers (PAs), which rely on EM mode confinement and resonant field enhancement. Interference-driven mechanisms, on the other hand, rely on the destructive or constructive interference between multiple resonant modes, including Fano resonance, electromagnetically induced transparency (EIT), electromagnetically induced absorption (EIA), absorption-induced transparency (AIT), and quasi-bound states in the continuum (quasi-BIC). These mechanisms typically exhibit ultra-narrow linewidths, high-Q, and superior S for trace analyte detection. These approaches enable ultrasensitive detection by transducing subtle refractive index changes, induced by analytes (e.g., biomolecules), into measurable shifts in frequency, amplitude, or phase. In this section, we discuss the physical principles of their implementation as well as the advantages and challenges of different approaches to guide us towards high-performance THz metasensor design.

2.1. Resonance-Driven

2.1.1. Surface Plasmon Polaritons

SPPs are hybrid EM waves that propagate along the boundary between a metal and a dielectric, generated by the coupling of THz radiation with collective electron oscillations at metal-dielectric interfaces.^[81–84] The strong localization of SPPs at the interface results in significantly enhanced near-field effects that are highly sensitive to changes in the surrounding refractive index. This characteristic renders SPP-based THz metasensors particularly effective for biosensing and chemical detection.

SPP-based THz metasensors are typically implemented via periodic metallic nanostructures, such as gratings, nanohole arrays, or plasmonic waveguides, which enable the excitation of surface plasmon waves. However, metals at THz frequencies exhibit significant ohmic losses, which will reduce the efficiency of SPP propagation. To overcome this constraint, researchers have proposed alternative solutions such as graphene-based SPPs or spoof SPPs on structured metals to mitigate losses and enable flexible, low-cost biosensors for label-free detection of biomolecules.

2.1.2. Localized Surface Plasmons

Unlike propagating SPPs, LSPs are confined resonances supported by subwavelength metallic nanostructures.^[85–87] LSPs exhibit ultra-high field enhancement in the nanogaps or tips, making them especially suitable for detecting trace molecules via refractive index shifts or molecular absorption enhancement. LSP-based THz metasensors are advantageous in detecting low-concentration analytes, including virus envelope proteins and small-molecule drugs, often enhanced further with metallic nanoparticles like Au nanoparticles (AuNPs) or Ag nanoparticles (AgNPs).

2.1.3. Mie Resonance

Mie resonance occurs in high-index dielectric particles (e.g., Si, Ge) when their dimensions are comparable to the incident wavelength, resulting in the formation of strong electric dipole (ED) and magnetic dipole (MD) resonances.^[88–91] In contrast to plasmonic structures, which suffer from ohmic losses, dielectric metasurfaces supporting Mie resonances demonstrate minimal loss and high-Q resonances. Meanwhile, their compatibility with non-metallic metasurfaces makes them ideal for flexible or implantable sensors.

THz metasensors based on Mie resonance are typically implemented using silicon or germanium metasurfaces, which support strong dipole and quadrupole resonances. These metasurfaces can be functionalized with molecular recognition elements to enable selective detection of biomolecules or chemical species. However, the Mie resonance conditions are strongly dependent on the size and refractive index of the dielectric particles, requiring precise nanofabrication.

2.1.4. Toroidal Resonance

Toroidal resonance is a distinct EM phenomenon arising from toroidal dipole moments, which are generated by circulating EM currents.^[92–94] Unlike conventional dipole or quadrupole resonances, toroidal modes exhibit unique near-field distributions that result in high-Q resonances with minimal radiative losses, making them exceptionally effective for trace analyte detection.

Although the mechanism offers remarkable advantages, its practical implementation encounters structural challenges. The excitation of toroidal modes typically relies on intricate 3D nanostructures that require engineered ring-like shapes to maintain toroidal dipole oscillations. These complex geometries, demanding precise alignment of 3D nanostructures, make fabrication more difficult compared to conventional resonant mechanisms.

2.1.5. Perfect Absorber

The PA is engineered to achieve complete absorption of incident THz waves through impedance-matched multilayered structures (e.g., metal-insulator-metal configurations), ensuring that all incoming energy is coupled into the structure.^[95–97] THz metasensors based on PA eliminate the reflection and transmission to

maximize energy dissipation in analyte-loaded regions, achieving low LoD and high S. This property makes them ideal for detecting trace quantities of biomolecules.

The realization of PAs-based THz metasensors involves the design of resonant structures, such as split-ring resonators (SRRs) to absorb THz waves at required frequencies. However, achieving perfect absorption often necessitates complex multi-layered configurations, which will increase the difficulty and complexity of manufacturing.

2.2. Interference-Driven

2.2.1. Fano Resonance

Fano resonance in THz metasensors can be realized by interference among different excitation modes, such as the interference between a broad continuum state and a narrow discrete state, yielding a highly asymmetric spectral profile.^[98–100] This distinctive, sharp spectral characteristic enhances S to minute refractive index variations, enabling accurate detection of molecular interactions, making it an efficient approach for THz metasensing.

In THz metasensors, Fano resonance is typically engineered with asymmetric SRRs or coupled plasmonic-dielectric structures. However, the asymmetric nature of the resonance requires exhaustive structural design to optimize S while minimizing the fabrication error to guarantee the performance of the metasensor.

2.2.2. Electromagnetically Induced Transparency

EIT is a quantum interference phenomenon that results in a narrow transparency window within an otherwise opaque spectral region.^[101–103] This effect occurs due to destructive interference between a low-Q bright mode (which interacts strongly with the incident field) and a high-Q dark mode (which is weakly coupled to the external field), resulting in a narrowband high-Q resonance. EIT-based THz metasensors can achieve frequency and phase sensing simultaneously. Compared with single-mode resonators, the steep phase gradient near the transparency window can significantly improve the FoM, enabling highly selective and sensitive detection of biomolecules.

In THz metasensing, EIT is achieved through coupled resonators, such as double SRRs or photonic crystal arrays. However, although EIT-based THz metasensors have exceptional S owing to their narrow spectral linewidth and strong near-field enhancement, maintaining the required interference conditions for EIT poses difficulties, as the balance between bright and dark modes is very sensitive to fabrication imperfections.

2.2.3. Electromagnetically Induced Absorption

EIA is the complementary phenomenon of EIT, where constructive interference results in a sharp absorption peak within a transmission background.^[104–106] This mode enhances analyte absorption fingerprint signatures, especially useful for molecular fingerprinting (e.g., amino acids, glucose) in aqueous or biological

environments. EIA-based THz metasensors allow for enhanced real-time, label-free spectroscopic analysis, even in the presence of complex background noise.

2.2.4. Absorption-Induced Transparency

AIT is a spectral interference phenomenon that occurs due to the coupling between a broadband resonance mode and a narrow-band molecular vibrational absorption.^[107–108] Unlike EIT, which results from interference between bright and dark EM modes, AIT arises from destructive interference between the broadband continuum of a metasurface and a molecular absorption line. This leads to a sharp transparency window within a broadband absorption profile.

AIT is particularly suitable for trace chemical sensing and fingerprint recognition of specific molecules (e.g., lactose, pesticides), especially in scenarios where characteristic absorption signatures are present. Compared with EIT approaches, AIT-based THz metasensors do not require engineered dual-mode structures, making it simpler for certain applications, albeit with lower Q.

2.2.5. Quasi-Bound States in the Continuum

BICs are localized nonradiative states with infinite Q, existing in the radiative continuum and capable of effectively trapping energy within the superstructure. However, in practical applications, infinite Q is difficult to achieve, so researchers usually introduce small perturbations to break the symmetry of the ideal BIC to excite quasi-BIC modes.^[109–111] Quasi-BIC-based THz metasensors can provide remarkable spectrum resolution, facilitating the detection of minuscule refractive index changes.

These metasensors are often realized with asymmetric periodic dielectric or hybrid nanostructures. Their robustness and compatibility with on-chip integration make them promising for clinical and point-of-care applications. However, the fabrication of quasi-BIC structures necessitates precise control of structural symmetry, hence challenging the manufacturing process.

3. Metrics

In this section, we introduce several key performance metrics, such as S, Q, FoM, and LoD, that are commonly used to evaluate and compare the sensing capabilities of THz metasensors, which serve as the foundation for performance analysis and future optimization.

3.1. Sensitivity

Generally, the S of a sensor refers to its ability to respond to changes in the external physical or chemical environment and is usually defined as the ratio of the change in the measured signal (ΔOutput) to the change in the input parameter (ΔInput), that is $S = \Delta\text{Output}/\Delta\text{Input}$. In THz metasensors, the definition of S depends on the sensing mechanism and the type of detection

signal. For the common frequency-sensitive sensor, S is the ratio of the resonance frequency changes (Δf) corresponding to the refractive index changes (Δn), which can be expressed as

$$S_f = \frac{\Delta f}{\Delta n} \quad (1)$$

Therefore, in the THz regime, the unit of S is THz per refractive index (THz/RIU). It is worth noting that for phase-sensitive sensors and amplitude-sensitive sensors, the calculation of S will change accordingly to $S_\varphi = \Delta\varphi/\Delta n$ and $S_I = \Delta I/\Delta n$, where $\Delta\varphi$ is the phase shift and ΔI is the intensity change.

3.2. Quality Factor

The Q-factor reflects the resonance characteristics of the sensors, which is generally defined by the ratio of the center frequency (f_0) to the full width at half maximum (FWHM) of the resonance peak:

$$Q = \frac{f_0}{\text{FWHM}} \quad (2)$$

It is worth noting that the higher the Q value, the higher the resolution and the lower the noise value, but high Q does not mean a high S, which depends on whether the mostly confined EM fields are fully overlapped with the analyte.^[12]

3.3. Figure of Merit

The FoM provides a trade-off between detection accuracy and spectral sharpness, which is defined by the ratio of the S to the FWHM,

$$\text{FoM} = \frac{S}{\text{FWHM}} \quad (3)$$

FoM is typically used to evaluate the actual sensor's performance. In some literature, FoM can also be calculated as $\text{FoM} = S \times Q$.

3.4. Limit of Detection

At last, LoD is the minimum concentration of an analyte that the sensor can reliably detect, where lower values correspond to better performance. Analytical LoD can be calculated as three times the ratio of the slope of the linear calibration/fit curve (k) to the standard deviation (SD) of the blank response,

$$\text{LoD} = \frac{3 \times \text{SD}}{k} \quad (4)$$

4. Applications of THz Metasensors

The ability of THz metasurfaces to manipulate EM waves on a subwavelength scale enables enhanced detection of biological, chemical, and physical properties with unprecedented precision,

making THz metasensor a powerful platform for highly sensitive, label-free detection and real-time monitoring in a variety of scientific and industrial fields. This section reviews the recent advances in THz metasensors in different sensing applications, including biological sensing, chemical sensing, material sensing, physical parameter sensing, high precision spectroscopy, etc.

4.1. Biological Sensing

THz metasensors have been widely employed in biosensing applications, providing a non-invasive and highly selective approach for protein, biomolecule detection.^[89,91,110,112–115,116–123] As illustrated in **Figure 2a**, Shi et al. (2022) developed an all-dielectric metasensor functionalized with AuNPs for highly sensitive and specific detection of hemagglutinin antigen at nanomolar concentrations.^[89] The metasensor, featuring a split half-cylinder array shown in **Figure 2a(i)**, provided strong field localization and an increased effective contact area, while AuNPs modified with anti-hemagglutinin tag antibodies enable specific immunobinding of hemagglutinin antigens. Experimental results demonstrated a 2.66-fold improvement in S ($2.96 \text{ GHz} \cdot \text{ml nmol}^{-1}$) and a detection accuracy of $1.05 \text{ nmol mL}^{-1}$ compared to the sensor without AuNPs. As shown in **Figure 2a(i)**, this sensor also exhibited high specificity to distinguish hemagglutinin antigen from nonspecific proteins (bovine serum albumin, ovalbumin, and whey protein) with significantly high- S . This work established a reliable platform for THz immunosensing, offering a promising approach for efficient, low-cost, and real-time biomolecular detection. In the same year, Yao et al. (2022) proposed a metasensor that effectively combines metal oxide semiconductor-like structures based on patterned graphene-polyimide-perovskite (MAPbI_3) with EIT-like metasurfaces to realize ultra-sensitive, multi-band, and multi-dimensional (frequency, amplitude, phase) detection of biological macromolecules, as shown in **Figure 2b**.^[118] The schematic layout of the proposed metasensor is depicted in **Figure 2b(i)**. They dropped five aqueous solutions of whey protein at various concentrations onto the surface of the metasensor to investigate the sensing performance. Experimental results showed that this sensor can detect whey protein at concentrations as low as 6.25 ng/mL , which is shown in **Figure 2b(ii)**. Meanwhile, the variations in transmission value, frequency shift, and phase difference increased with whey protein concentration, reaching 82%, 400 GHz, and 360° , indicating that the proposed metasensor can detect multi-dimensional dimensions as valuable sensing information, which holds significant potential for the application of THz metasensors in biosensing. Two years later, Zhang et al. (2024) proposed a metasensor for the specific detection of serum amyloid A (SAA) antigen in femtomolar concentration, as shown in **Figure 2c**.^[123] Their metasensor supports a Fano resonance mode at 0.73 THz, characterized by anti-parallel surface currents and weak radiation coupling, leading to a high Q-factor and narrow spectral linewidth, which makes the resonance highly sensitive to tiny dielectric perturbations caused by a few molecules within the hotspot region. Furthermore, by modifying the single-walled carbon nanotube (SWCNT) film with concentrated sulfuric acid and introducing functionalized AuNPs, the metasensor significantly

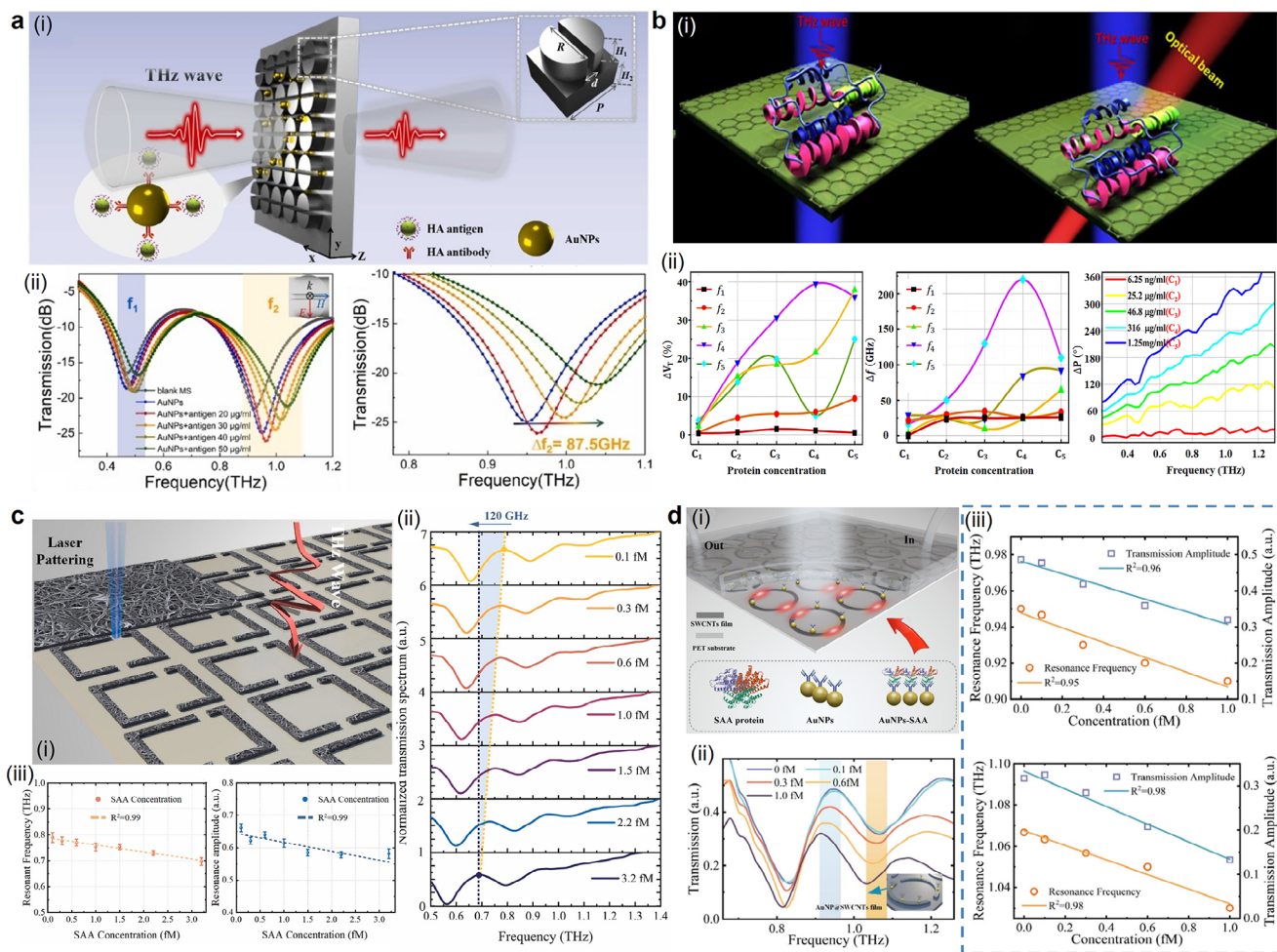


Figure 2. a-i) Schematic of the all-dielectric metasensor consists of a split half-cylinder array, which is illuminated by a normally incident THz wave. ii) Experimental transmission spectra and the enlarged view of the frequency shifts of y-polarization for hemagglutinin antigen from 20 to 50 $\mu\text{g mL}^{-1}$ on metasurface sensors with the functionalized AuNPs. b-i) Schematic of the proposed metasensor. ii) Quantitative analysis of multi-dimensional sensing performance. The transmission value difference ΔV_T , frequency shift Δf , and phase difference ΔP between the bare sensor and different concentrations of whey protein at frequencies f_1 – f_5 . c-i) Schematic diagram of the THz metasensor structure. ii) Normalized transmission spectra of the THz metasensor covered with SAA protein at different concentrations. iii) The resonance frequency shift and the formant amplitude changes with the sample concentration, and the dotted line represents function fitting is linear with the experimental data. d-i) Schematic of the unlabeled THz metasensor designed for the selective capture of SAA protein with the incorporation of functionalized AuNPs. ii) Transmission spectrum of the THz metasensor based on AuNPs film changes with SAA protein concentration. iii) the changes of metasensor transmission frequency shift and resonance transmission amplitude at the lower and higher resonance frequency positions with concentration. a) Reproduced with permission.^[89] Copyright 2022, Elsevier. b) Reproduced with permission.^[118] Copyright 2022, De Gruyter. c) Reproduced with permission.^[123] Copyright 2024, Elsevier. d) Reproduced with permission.^[91] Copyright 2025, Elsevier.

enhanced the binding specificity toward a small number of analyte molecules, enabling their presence to induce a measurable resonance frequency shift. Meanwhile, the acid treatment effectively reduced the non-resonant loss of the SWCNT film. Benefiting from the high specific surface area of SWCNTs, the adsorption capability for target biomolecules was further improved, thereby facilitating highly selective detection of trace amounts of SAA protein. The schematic diagram of the designed metasensor is shown in Figure 2c(i), which consists of a SWCNT film with asymmetric split ring units on a polyethylene terephthalate substrate. As shown in Figure 2c(ii), seven SAA solutions (0.1–3.2 fM) were tested, and the resonance frequency of SAA protein at 3.2 fM shifted by 120 GHz, denoting the S of this THz

metasensor for SAA detection is 37.5 GHz fM⁻¹. Meanwhile, the frequency shift and amplitude modulation as functions of SAA concentration were calculated for the quantitative analysis of the detection characteristics of this metasensor. As shown in Figure 2c(iii), there is a clear linear shift trend in both resonance frequency and amplitude as concentration increases, signifying the sensor's stable and reliable performance. The extreme S at the fM level is a combined result of resonator physics (Fano resonance), material engineering (low-loss SWCNTs), and molecular capture (functionalized AuNPs). Together, they allow even a few molecules in a small ($\approx 100 \mu\text{m} \times 100 \mu\text{m} \times 10 \mu\text{m}$ for a unit cell) volume to be reliably detected through significant spectral modulation. Most biomolecules in clinical

applications require a liquid environment to maintain their activity; however, the substantial absorption characteristics of water for THz waves significantly restrict the THz sensor's detection performance. To address this challenge, Zhang et al. (2025) developed a high-performance THz metasensor, which is composed of periodically arranged asymmetric open resonant rings, for the highly sensitive direct detection of trace SAA protein solutions in a microfluidic platform, as illustrated in Figure 2d(i).^[91] This metasensor, integrated with microfluidic channels, can directly detect SAA proteins in solution at the fM level. Figure 2d(ii,iii) show the experimental results with the concentration gradient of the SAA protein established at 0.1, 0.3, 0.6, and 1.0 fM, demonstrating that the frequency sensing S of the metasensor at the lower resonance position can achieve 41 GHz fM^{-1} while the resonance amplitude S is 0.20 fM^{-1} and the linear fitting goodness of fit (R^2) is 0.95 and 0.96, respectively. This method not only advanced the design and functionality of a novel highly sensitive carbon-based THz metasensor but also established a basis for its application in the detection of trace biological material in liquid environments.

Besides detecting proteins and biomolecules, THz metasensors can also be extensively utilized for the detection of pathogens, microorganisms, chiral compounds, and drugs.^[84,107,124–131] As shown in Figure 3a, Jun et al. (2022) developed a label-free metasensor for identifying pathogens based on thermal analysis.^[126] They utilized the inherent characteristic that the dielectric constant of the bacterial layer varies with temperature and obtained the differential thermal curve by monitoring the correlation between the resonant frequency and temperature, whose peaks were consistent with the phase transition during the growth and death of the individual cells to offer unique fingerprints specific to individual microbes, as shown in Figure 3a(i,ii). Furthermore, as illustrated in Figure 3a(iii), they conducted THz thermal curve analysis on mixed bacterial species to illustrate the capability of the engineered metasensor to differentiate non-pathogenic bacteria from pathogenic bacteria, which is essential for practical applications. One year later, Lyu et al. (2023) designed a frequency-selective fingerprint sensor with cross-slot unit cells to improve THz molecule trace detection by AIT and multiplexing technologies.^[107] Figure 3b(i) shows a sensor consisting of a 6×6 -pixel array, with each pixel (P01–P36) measuring 3×3 mm and arranged according to different lengths of the cross-slot unit cell. The absorption characteristics of D-carnitine and L-carnitine were determined by calculating the absorbance signal, as illustrated in Figure 3b(ii), which exhibits significant enhancement compared to the absorbance characteristic signal (red line) without the sensor. Such two absorption properties distinctly indicate the unique features of D-carnitine and L-carnitine. The retrieved absorbance envelope amplifies the peak values to 24.1% and 30.3% (D-carnitine) and 30.9% and 25.2% (L-carnitine) at the frequencies of 1.39 and 1.72 THz, and the mean enhancement factors for D-carnitine and L-carnitine are around 8.3 dB (6.8 times) and 8.65 dB (7.3 times) within the frequency range of 0.95–2.0 THz, respectively. The flexibility and versatility of the proposed sensor may have potential applications in boosting trace chiral enantiomer detection and expanding molecular fingerprint recognition through multiplexing and AIT. Lou et al. (2024) introduced a metasensor with the unit cell of two

SRRs positioned on both sides of the central cut-wire, as shown in Figure 3c.^[129] When the electric and toroidal dipoles with identical intensity radiate in antiphase on the metasensor, the quasi-anapole mode will be induced to realize a high Q factor and FOM, as illustrated in Figure 3c(i). In practical application, they conjugated the functionalized gold nanoparticles with the specific monoclonal antibody onto the metasensor to detect the cell concentration of spike protein. The experimental results indicate that when the concentration of anti-spike is 50 pg mL^{-1} , the LoD reaches 3 pg/mL (39 fmol), which is shown in Figure 3c(ii). In the same year, Zhang et al. (2025) proposed a data iterative strategy based on deep learning to achieve the customized design of chiral metasurfaces for precisely identifying different chiral molecules in a label-free manner, as shown in Figure 3d(i).^[131] Compared with traditional methodologies that predominantly depend on simulations to obtain comprehensive data, this strategy offers maximized degrees of freedom without incurring excessive time costs. Figures 3d(ii,iii) illustrate the evaluation of the customized metasurface's S and selectivity for chiral molecule detection, wherein glutamate enantiomers were chosen, and the circular dichroism (CD) spectra and angle differences of L- and D-glutamate were experimentally assessed at concentrations of 0.1 and 0.3 mg mL^{-1} , respectively. The maximum CD angular differences are 32.50° and -12.47° at concentrations of 0.3 and 0.1 mg mL^{-1} , respectively, demonstrating the chiral sensing capability of the proposed chiral metasurface to differentiate between L- and D-chiral enantiomers.

4.2. Chemical Sensing

In this section, we explore recent advancements in THz metasensors for chemical and environmental sensing, highlighting their applications in detecting plant-growth-regulator residues, citrate salts, pesticides, and fluoroquinolone antibiotics with high- S , rapid detection, and integration with machine learning for precise analysis.^[132–141] As illustrated in Figure 4a, Du et al. (2021) proposed a highly sensitive double-formant metasensor for detecting plant-growth-regulator residues.^[133] Figure 4a(i) depicts the image of the designed metasensor, which consists of meta-atom arrays with two rings with four gaps. The meta-atom was composed of three layers: polyimide, titanium, and gold. Experimental results indicated that the platform could rapidly detect plant growth-regulators in aqueous solutions at concentrations as low as 0.05 mg mL^{-1} , which is shown in Figure 4a(ii). This approach offers a fast, low-cost, and highly sensitive solution for plant-growth-regulator residue detection, with significant potential for applications in food safety and THz sensing. Deng et al. (2022) presented a THz metasensor with an asymmetric double-opening ring structure for the rapid and precise detection of citrate salts in food and pharmaceuticals, as shown in Figure 4b(i).^[135] Through the optimization of the sensor's design and analyzing factors such as opening positions, meta-atom arrangement, and analyte properties, the sensor demonstrated high- S to refractive index changes. Finally, six low-concentration citrate salt solutions were successfully identified, showcasing the potential of THz spectroscopy combined with metamaterials for chemical sensing, as illustrated in Figure 4b(ii). This

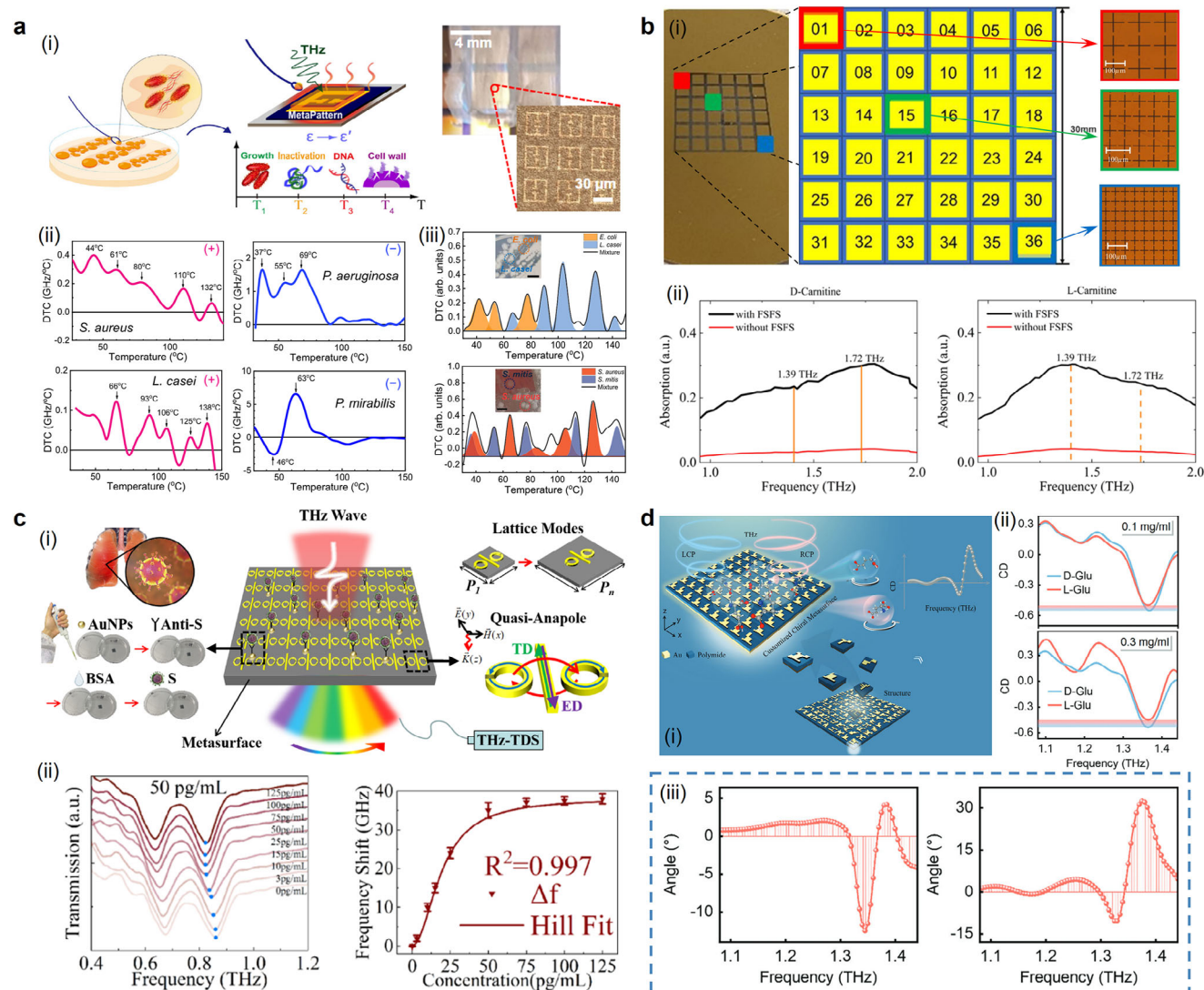


Figure 3. a-i) Schematic of experiments and photograph of metasensor coated with a microbial film. ii) Differential thermal curve results for four bacterial species (*S. aureus*, *P. aeruginosa*, *L. casei*, and *P. mirabilis*). iii) Thermal curve analysis for a mixture of bacteria. The top is a differential thermal curve for a mixture of *E. coli* and *L. casei* bacteria, and the bottom is a differential thermal curve for a mixture of *S. aureus* and *S. mitis* bacteria. b-i) Optical images of the fabricated frequency-selective fingerprint sensor with 36-pixel location (P01-P36). ii) Measured envelope absorbance signals of coating 10 μm thick (Left D-carnitine, Right L-carnitine) on the frequency selective fingerprint sensor. c-i) The schematic of the sensing strategy based on the proposed metasensor. ii) Transmission spectra with antibody concentration of 50 pg mL^{-1} and corresponding frequency shift vs spike protein concentrations. d-i) Schematic illustration of chiral metasurfaces that demonstrate distinct absorption behaviors when subjected to circularly polarized incidence with opposite spins. ii, iii) CD spectra and angular differences were obtained for L- and D-glutamic acid at concentrations of 0.1 and 0.3 mg mL^{-1} . a) Reproduced with permission.^[126] Copyright 2022, Springer Nature. b) Reproduced with permission.^[107] Copyright 2023, Springer Nature. c) Reproduced with permission.^[129] Copyright 2024, Elsevier. d) Reproduced with permission.^[131] Copyright 2025, John Wiley and Sons.

method offers a promising solution for quality control and safety monitoring in the food and pharmaceutical industries. Wang et al. (2024) presented a dual-band flexible THz metasensor, fabricated via ultraviolet (UV) lithography and characterized using a THz time-domain spectroscopy system, exhibiting resonances at 0.37 and 1.13 THz, as illustrated in Figure 4c.^[139] The schematic diagram of the proposed metasensor is shown in Figure 4c(i), which consists of two layers: a metal periodically patterned array on the top layer and a substrate layer on the bottom layer. The unit cell consisted of four identical spiral-shaped

strips stacked in a rotational orientation. It excels in pesticide detection, with graphene-coated surfaces enhancing S for low-concentration chlorpyrifos-methyl solutions due to strong $\pi-\pi$ stacking interactions, while the uncoated sensor is preferred for high-concentration detection, which is shown in Figure 4c(ii). This work advances THz sensing technology, offering a versatile platform for pesticide detection across a broad concentration range. Zhang et al. (2025) proposed a novel method for detecting trace fluoroquinolone antibiotics utilizing THz metasensors in conjunction with machine learning algorithms.^[141] As

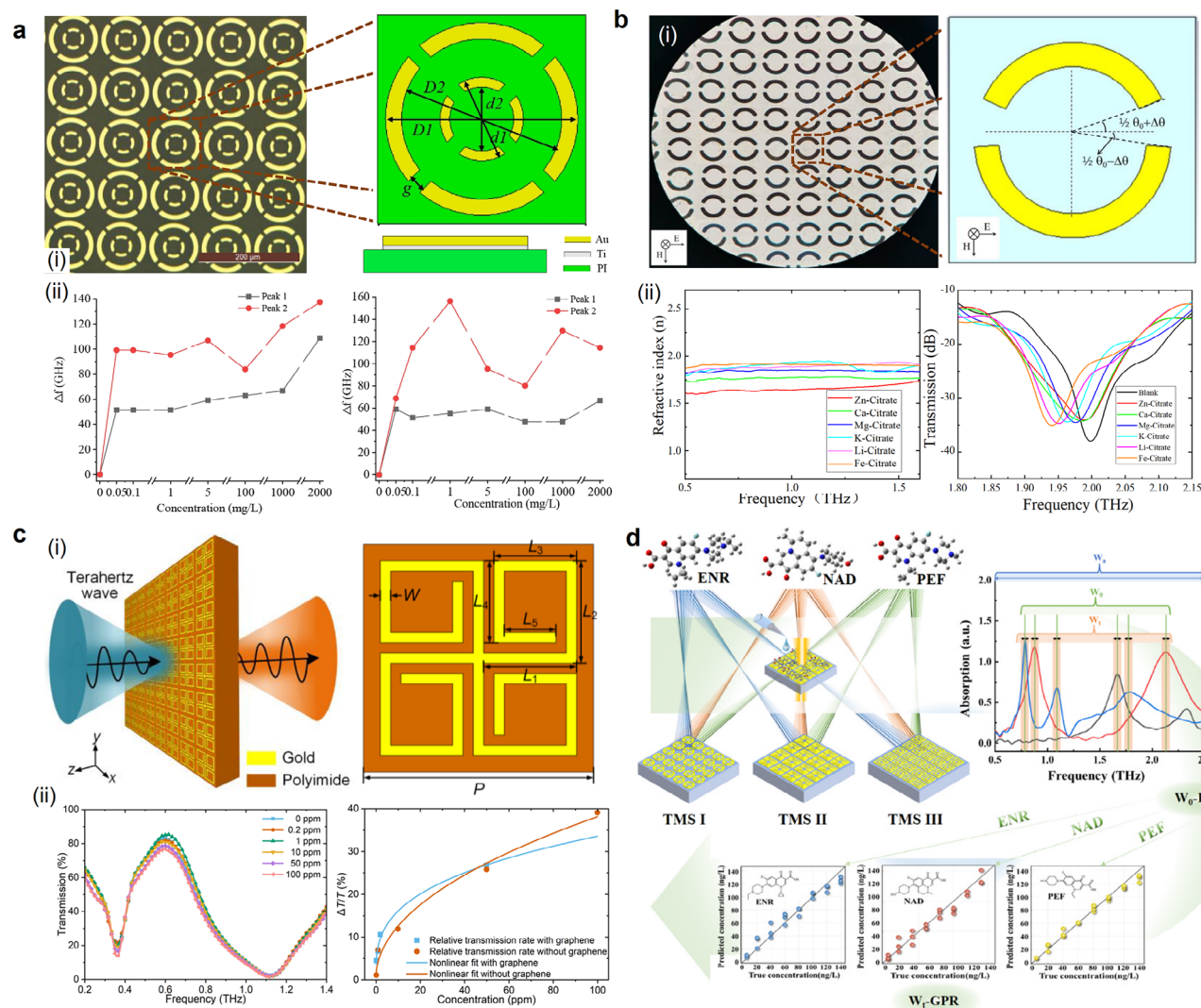


Figure 4. a-i) Partial image of the designed metasensor and schematic diagram of a unit cell. ii) The resonant peaks shift for different butyl hydrazine and N-N diglycine concentrations. b-i) Partial image of the designed metasensor and schematic diagram of a unit cell. ii) Refractive indices of six different citrate salts and transmission spectra of the blank sensor and the sensor covered with six different citrate salts. c-i) Schematic of the proposed metasensor and unit cell. ii) Transmission spectrum of the metasensor with the pesticide concentration increasing from 0 to 100 ppm (left) and relative transmission rates of the metasensor with and without coated graphene, as the chlorpyrifos-methyl solution rises from 0 to 100 ppm (right). d-i) Schematic diagram and the design method of the proposed metasensor, and quantitative results of the w_0 -GPR model with different kernel functions for three trace fluoroquinolone antibiotics. a) Reproduced with permission.^[133] Copyright 2021, The Optical Society. b) Reproduced with permission.^[135] Copyright 2022, Multidisciplinary Digital Publishing Institute. c) Reproduced with permission.^[139] Copyright 2024, American Chemical Society. d) Reproduced with permission.^[141] Copyright 2025, Elsevier.

shown in Figure 4d, three patterned metasensors were designed and fabricated to capture distinct resonance responses for fluoroquinolone antibiotics, specifically enrofloxacin (ENR), nadifloxacin (NAD), and pefloxacin (PEF). They constructed a resonant peak feature matrix (w_0) and incorporated it with the K-Nearest Neighbor (KNN) algorithm to achieve 100% classification accuracy. Additionally, an optimal resonant peak interval feature matrix (w_i) was created and integrated with Gaussian Process Regression (GPR) for quantitative analysis, achieving high accuracy ($R^2 = 0.94$ – 0.98) and stability. This approach enables rapid, precise, and reliable detection of trace fluoroquinolone antibiotics, offering significant potential for applications in food safety and environmental monitoring.

4.3. Material Sensing

The resonance characteristics of THz metastructures are highly sensitive to changes in the surrounding material. This highly responsive characteristic makes the THz metasensor well-suited for sensing nanometer-scale film materials, which are more than 1000 times smaller than the THz wavelength.^[142–146] For example, in the Figure 5a, Jáuregui-López et al. (2018) showed the thickness sensing of the analyte using a hole array pattern that supports extraordinary optical transmission resonance.^[144] The hole diameter is $a = 105 \mu\text{m}$, and the polypropylene substrate has thicknesses of 50 and 75 μm . In this research, it was observed that sensing performance is enhanced when the ana-

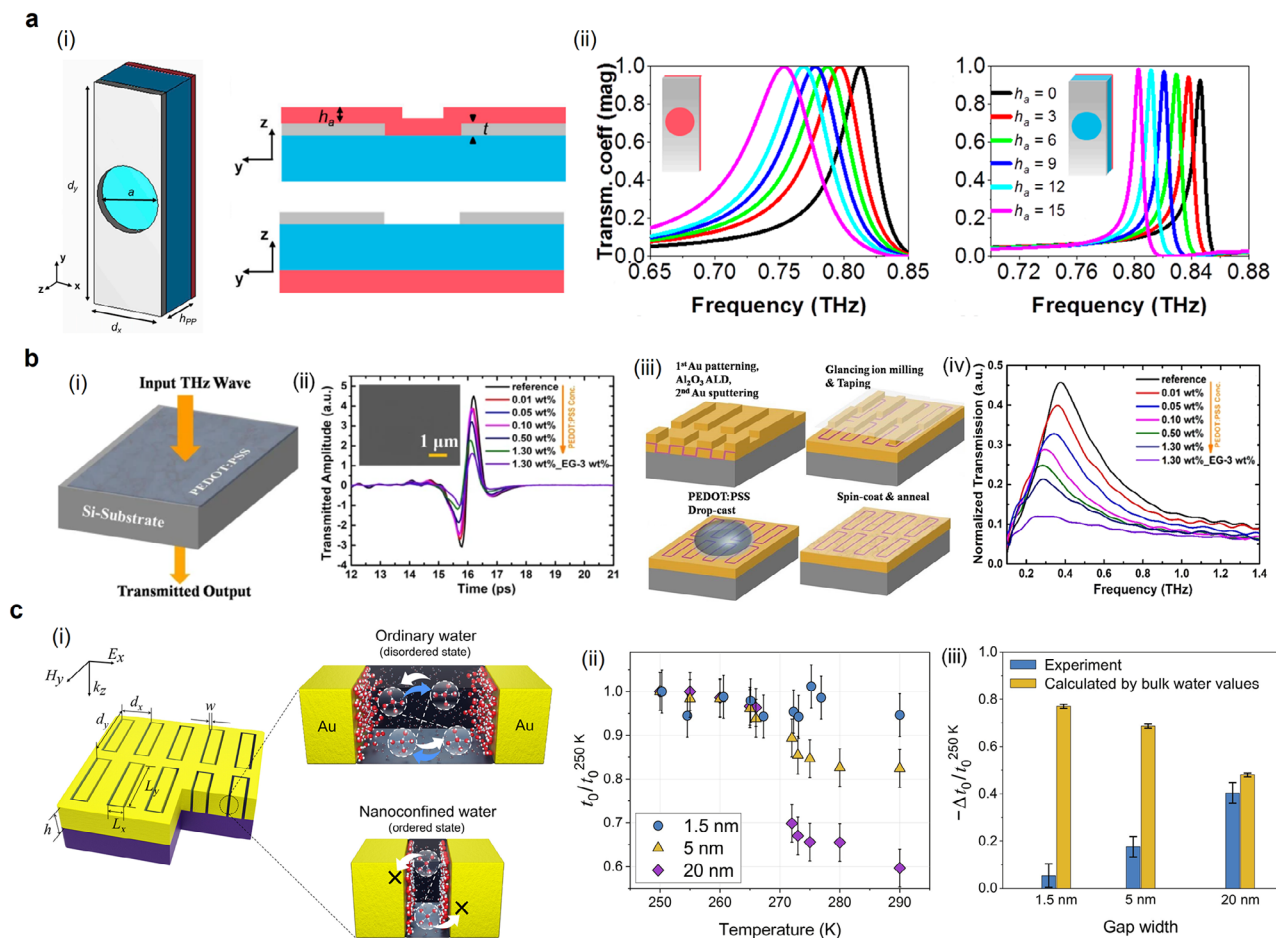


Figure 5. a-i) Schematic of the hole array metasensor. Blue and red layers are the polypropylene substrate and the analyte, respectively. ii) Changes in the THz transmission spectrum depending on the position and thickness (h_a) of the analyte. b-i) Schematic diagram of PEDOT: PSS-coated Si substrate. ii) Time-domain spectrum changes of the THz transmission according to the concentrations of PEDOT:PSS. iii) Schematic diagram of the coating process of PEDOT:PSS on the nanogap metamaterial. iv) THz transmission spectrum for coated nanogap metamaterial with different wt% of PEDOT:PSS. c-i) Schematic of the water-confined nanogap metamaterial with different gap widths. ii) The relative THz transmission at the resonance peak position, depending on the temperature, for different gap widths. iii) Comparison of the relative peak amplitude of experimental and calculated results with different gap widths. a) Reproduced with permission.^[144] Copyright 2018, MDPI. b) Reproduced with permission.^[145] Copyright 2019, Springer Nature. c) Reproduced with permission.^[146] Copyright 2024, American Association for the Advancement of Science.

lyte film is deposited on the non-patterned side. This configuration offers the advantage of allowing the structure to be cleaned without damaging the metal pattern in routine applications. Chaudhary et al. (2019) conducted the THz sensing based on the doping concentration of the PEDOT:PSS film as shown in Figure 5b.^[145] When the PEDOT:PSS film is coated on a THz nanogap metamaterial (Figure 5a(iii)), it exhibits significantly higher S compared to when it is coated on a Si substrate only (Figure 5b(i)). The THz nanogap metamaterial was designed with a 20 nm gap along a loop measuring $20 \times 80 \mu\text{m}$. This nanometer-scale gap was highly suitable for material sensing applications because it can enhance the THz wave in the near-field by three orders of magnitude or even more. These works demonstrate that THz metasensors are highly effective for analyzing the properties of nanometer-thin films, which are significantly smaller than the THz wavelength. Furthermore, the integration of functional materials with the THz metasensor offers new possibilities for photonic device applications. On the

other hand, the THz region responds to the collective dynamics of molecules due to its low energy. This can be utilized for analyzing the collective dynamics of water molecules, and the THz metasensor can provide a more sensitive response. Yang et al. (2024) analyzed the dynamics of water confined in the gap region of the nanogap metamaterial (Figure 5c).^[146] Interestingly, as the gap size decreased, an ice-like feature was observed, which is completely different from the expected transmission spectrum trend in the water phase. This methodology offers an unprecedented approach to studying and utilizing water-mediated processes such as protein folding, lipid rafts, and molecular recognition.

4.4. Physical Sensing

The strong coupling between the THz metastructure and surrounding materials enables easier detection of the intrinsic

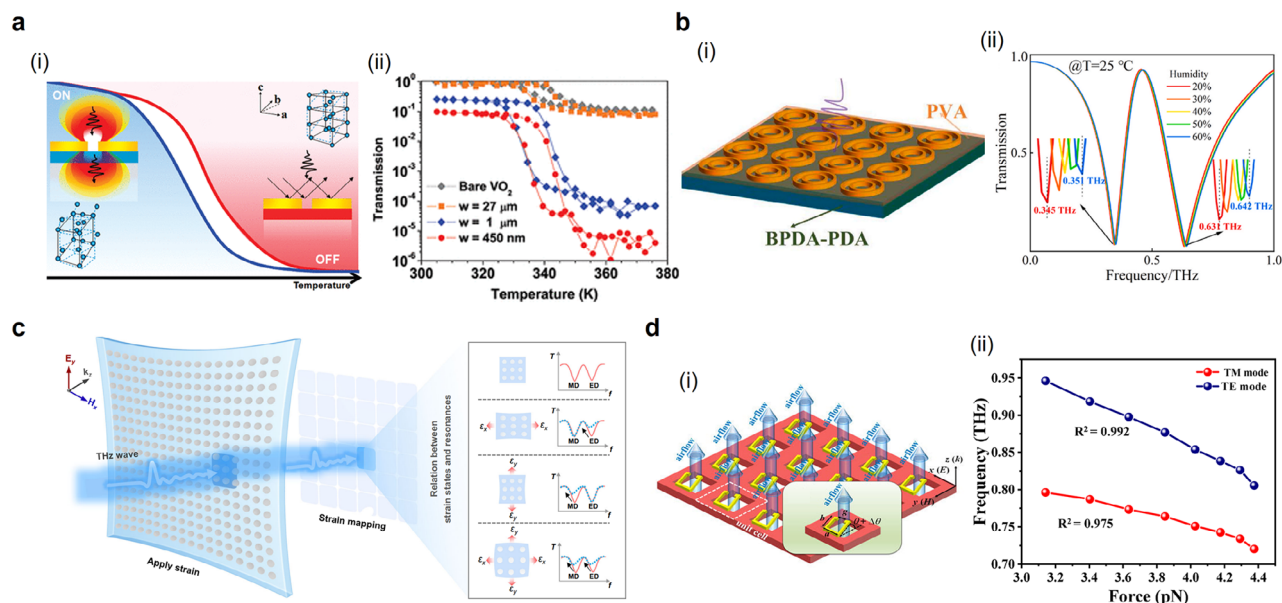


Figure 6. a-i) Phase transition diagram of metamaterial on VO₂ thin film. ii) THz transmission hysteresis curve as a function of temperature for bare VO₂ film and metamaterial on VO₂ film at 0.4 THz. b-i) Schematic of the THz humidity metasensor. ii) Resonance frequency shift of the metasensor according to humidity variations. c) Schematic diagram of the THz strain metasensor and spectral change according to strain variations. d-i) Schematic of the MEMS-based THz pressure and flow metasensor. ii) Resonance frequency changes according to the applied external force, by the airflow rate. a) Reproduced with permission.^[147] Copyright 2010, American Chemical Society. b) Reproduced with permission.^[154] Copyright 2024, Elsevier. c) Reproduced with permission.^[155] Copyright 2024, Springer Nature. d) Reproduced with permission.^[156] Copyright 2024, MDPI.

physical properties of the material.^[147–149] For example, Seo et al. (2010) integrated the THz metastructure with a VO₂ film, which exhibits phase changes with temperature (Figure 6a(i)).^[147] As shown in Figure 6a(ii), the VO₂ film coupled with the THz metastructure demonstrated a significantly greater modulation depth compared to bare VO₂. Additionally, the increased S of the metasensor, achieved through a smaller gap width, further enhanced this effect. The introduction of the metasensor has led to a controllable dynamic range enhancement of VO₂ film by a factor of 1000, and up to 10 000. This VO₂-based THz metasensor can be utilized for bolometric applications that measure temperature. THz metamaterials can be utilized for sensing not only temperature but also other physical parameters such as humidity, strain, and pressure.^[150–156] Ma et al. (2024) showed a THz humidity sensor composed of a double golden ring fabricated on a p-Phenylene biphenyltetracarboximide substrate, combined with the hydrophilic material polyvinyl alcohol (Figure 6b(i)).^[154] This substrate is highly sensitive to temperature changes, while polyvinyl alcohol is sensitive to humidity. In this study, the changes in the material properties of the surrounding material of the golden ring due to humidity were captured by monitoring the resonant frequency shift, allowing for the detection of both temperature and humidity. By studying the relationship between the resonant frequencies at the two resonance dips and the environmental parameters, the relative humidity S was found to be 0.14 GHz/% at the low frequency (first dip) and 0.27 GHz/% at the high frequency (second dip). The temperature S was 1.35 GHz °C⁻¹ at the first dip and 2.45 GHz °C⁻¹ at the second dip. Moreover, it was demonstrated that the effects of temperature and relative humidity are independent of each other, proving that both relative humidity and temperature

changes can be detected simultaneously. Lu et al. (2024) reported a flexible metasensor array that can detect the in-plane direction and magnitude of pre-loaded strains by referencing transmitted THz signals based on Mie resonance (Figure 6c).^[155] This THz strain metasensor was composed of a ZrO₂ microsphere array on a polydimethylsiloxane (PDMS) substrate and was fabricated using a micro template-assisted assembly strategy. The deep decoupling feature of ED and MD resonances enables the differentiation of the direction of external deformation. This is related to the electric field distributions of the ED and MD resonances, and the non-interfering tuning of the two resonances was achieved by changing the gap between the microspheres. As shown in Figure 6d, Lai et al. (2023) demonstrated the MEMS-based THz pressure and flow metasensor.^[156] This metasensor utilized a patterned silicon substrate with a Through Si Via (TSV) process and a gold SRR. Here, the SRR was designed as a cantilever structure within the TSV. When the airflow passes through the TSV from bottom to top and bends the SRR cantilever, the SRR cantilever bends upwards. By bending the SRR cantilever, the EM response of the metamaterial was tuned, showing polarization-dependent characteristics. In this study, by changing the angle of the SRR cantilever from 10° to 45° depending on the airflow rate, the resonance can be modulated with tuning ranges of 0.075 THz in the transverse magnetic mode and 0.14 THz in the transverse electric mode.

4.5. Precision Spectroscopy

Various strategies are being introduced to further increase the S of sensors for precision spectroscopy. In Figure 7a, Xu et al.

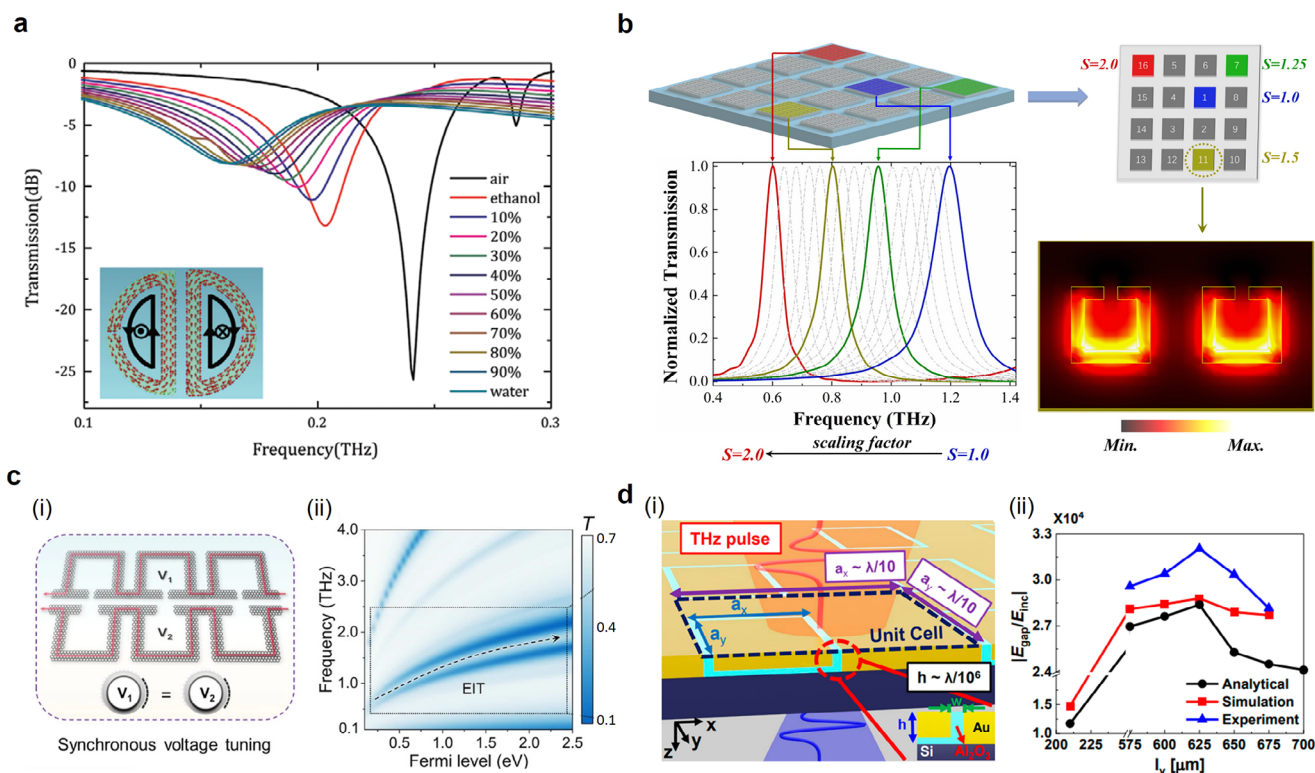


Figure 7. a) Simulated THz transmission spectra of a dual toroidal metasensor with different ethanol concentrations. b) Schematic of the pixelated metamaterials for molecular fingerprint detection. c-i) Schematic of the broadband tuning metasensor using EIT. ii) The broadband resonance frequency tuning of the metasensor according to the Fermi level of graphene. d-i) Schematic of the nanogap metamaterial for obtaining strong electric field enhancement in the gap region. ii) Comparison of the field enhancements between calculation and experimental results according to the y-periodicity change. a) Reproduced with permission.^[93] Copyright 2021, John Wiley and Sons. b) Reproduced with permission.^[165] Copyright 2024, Optica Publishing Group. c) Reproduced with permission.^[102] Copyright 2024, Springer Nature. d) Reproduced with permission.^[179] Copyright 2023, American Chemical Society.

(2021) showed a dual toroidal torus metasensor integrated with a microfluidic channel.^[93] In this study, the toroidal metasurface was fabricated by overlapping two mirrored SRRs to generate a stronger toroidal moment, achieving efficient detection of a mixed ethanol-water solution. Meanwhile, various strategies have been proposed for detecting THz molecular fingerprints.^[94,101,102,109,157–173] Li et al. (2024) fabricated a THz sensor that consists of a 4×4 pixelated filter-detector array (Figure 7b).^[165] The complementary SRR structure fabricated on a MgO substrate covers different resonance frequency ranges for each pixel in the 0.6–1.2 THz. The frequency difference between two adjacent pixels' resonance frequencies corresponded to the spectral resolution of 40 GHz. These pixelated metasensors can be used to analyze molecular absorption fingerprints, such as those of biomolecules, by correlating the transmission of each exposed meta-pixel with the molecular absorption at the resonance frequency of each meta-pixel. As shown in the Figure 7c, Liu et al. (2024) demonstrated a single-pixel reconfigurable graphene metasurface based on EIT for detecting molecular fingerprints.^[102] The Fermi level of graphene, which can be modulated via electrical gating, enables the tuning of the resonance frequency, and in this study, the 0.5–2 THz range tuning was demonstrated. Additionally, the metasurface successfully obtained the molecular fingerprint spectra of chiral optical isomers,

L-carnitine and D-carnitine. To enhance the sensing S in the THz range, design methods utilizing artificial intelligence technology have been employed.^[174–179] For example, as illustrated in the Figure 7d, Lee et al. (2023) optimized the parameters for the THz nanogap metamaterial using a reinforcement learning-based inverse design approach.^[179] In this study, an optimized nanogap metamaterial was experimentally fabricated on a silicon substrate, achieving an electric field enhancement of more than 30 000-fold in the 6G frequency range. This technology can be applied to the development of detectors with high- S to extremely weak THz electric fields, as well as for ultra-trace molecular detection. Moreover, recent advances in THz metasurfaces have focused on exploiting non-Hermitian physics by engineering higher-order and resonant exceptional points, where eigenvalues and eigenvectors coalesce.^[180–181] These exceptional points significantly enhance sensor S and resolution by amplifying the system's response to minute perturbations. Such non-Hermitian metasurface designs offer a promising pathway for further improving the performance of precision THz spectroscopy and sensing technologies.

To establish a clear relationship between the physical mechanisms discussed in Sections 2.1 and 2.2 and representative experimental implementations, we compiled a summary of various THz metasensors in Table 1. Each entry is categorized based

Table 1. Summary of THz meta-sensors.

Ref	Time	Application	Detection sample	Principles	S	Q-factor	FOM	LoD
[113]	2021	Biological sensing	Midkine	EIT + LSP	49 GHz	N/A	N/A	0.5 $\mu\text{g mL}^{-1}$
[114]	2021	Biological sensing	Bovine Serum Albumin	Toroidal resonance	72.81 GHz/(ng mm ⁻²)	4.03	2.476	0.035 mg mL ⁻¹
[115]	2021	Biological Sensing	Refractive index (1.5-2)	EIT	280.8 GHz RIU ⁻¹	14.3	4	N/A
[89]	2022	Biological sensing	HA antigen	Toroidal resonance	2.96 GHz·mL nmol ⁻¹	N/A	0.952	20 $\mu\text{g mL}^{-1}$
[116]	2022	Biological sensing	CA125 CA199	LSP	65 GHz RIU ⁻¹	N/A	N/A	0.01 U mL ⁻¹
[120]	2023	Biological sensing	A β protein	Toroidal resonance	258 GHz RIU ⁻¹	N/A	N/A	0.1 $\mu\text{g mL}^{-1}$
[121]	2023	Biological sensing	Casein protein	PA	251 GHz (frequency) 327° (phase)	N/A	3.54 ng mL ⁻¹	N/A
[110]	2023	Biological sensing	CRP protein SAA protein	Quasi-BIC	674 GHz RIU ⁻¹	50	N/A	1 pM
[122]	2024	Biological sensing	DOPC emulsion/solution	LSP	30 GHz	16.1	N/A	N/A
[123]	2024	Biological sensing	SAA protein	Fano resonance	37.5 GHz fM ⁻¹	N/A	N/A	0.1 fM
[91]	2025	Biological sensing	SAA protein	Fano resonance	41 GHz fM ⁻¹ (frequency) 0.20 fM ⁻¹ (amplitude)	N/A	N/A	0.1 fM
[128]	2024	Biological sensing	Yeast cells	PA	8.6×10^3 GHz· μm^2 cell ⁻¹	6.0	78 μm^2 cell ⁻¹	7×10^{-3} cell μm^{-2}
[132]	2021	Chemical sensing	Chlorothalonil	LSP	156 GHz RIU ⁻¹	3.6	0.38	N/A
[133]	2021	Chemical sensing	Butylhydrazine N-N diglycine	Fano resonance	156.43 GHz/(mg mL ⁻¹)	61.85	22.15	0.05 mg mL ⁻¹
[155]	2024	Physical sensing	2D strain	Mie resonance	0.4215 THz (x-direction) 0.3220 THz (y-direction)	N/A	N/A	1.25% (x) 2.7% (y)
[93]	2021	Precision spectroscopy	Ethanol	Toroidal resonance	124.3 GHz RIU ⁻¹	8.78	0.366	N/A
[160]	2024	Precision spectroscopy	Ethanol Sucrose Oil	PA	120.6 GHz RIU ⁻¹	82.3	20.1	N/A
[101]	2024	Precision spectroscopy	α -lactose Benzoic acid Vitamin B2 2,5-dichloroaniline	AIT	0.00211/(mg mL ⁻¹) 0.00208/(mg mL ⁻¹) 0.00211/(mg mL ⁻¹) 0.00219/(mg mL ⁻¹)	N/A	N/A	8.61 mg mL ⁻¹ 6.96 mg mL ⁻¹ 7.54 mg mL ⁻¹ 8.35 mg mL ⁻¹

on its underlying sensing principle, including EIT, Fano resonance, LSPs, PA, and quasi-BIC, thus establishing a direct correspondence between theoretical classification and practical implementation. In addition to the mechanism type, the table includes key performance indicators such as S, Q-factor, FOM, and LoD, enabling a quantitative comparison across diverse metasensing strategies. The comparative analysis offered by Table 1 also provides insight into the relative practicality and performance trade-offs among different mechanisms. For example, quasi-BIC and Fano resonance-based designs demonstrate the highest S (e.g., 674 GHz RIU⁻¹ for quasi-BIC, 156.4 GHz RIU⁻¹ for Fano resonance), highlighting their suitability for ultra-sensitive detection applications. PA-based metasensor, while exhibiting slightly lower S, offers notably high FOM values (e.g., 22.15 for Fano, 20.1 for PA), making them attractive for broadband and high-resolution spectral detection, such as in precision spectroscopy.

EIT-based metasurfaces offer the advantage of multi-band response with moderate design complexity, which makes them well-suited for refractive index-based chemical or biological sensing. By combining physical classification with performance data, Table 1 serves as a practical guide to evaluate the trade-offs and application relevance of different THz metasensing mechanisms.

Despite the promising performance shown in Table 1, the practical application of THz metasensors, especially in biological or aqueous environments, remains significantly constrained. A major challenge lies in the intrinsic absorption of THz radiation by water, which stems from its strong dipolar nature. This results in substantial signal attenuation and a reduced signal-to-noise ratio when probing liquid samples. Consequently, although THz metasurfaces offer considerable potential for high-S sensing, their effectiveness in biological and chemical detection involving aqueous media is often limited by these absorption

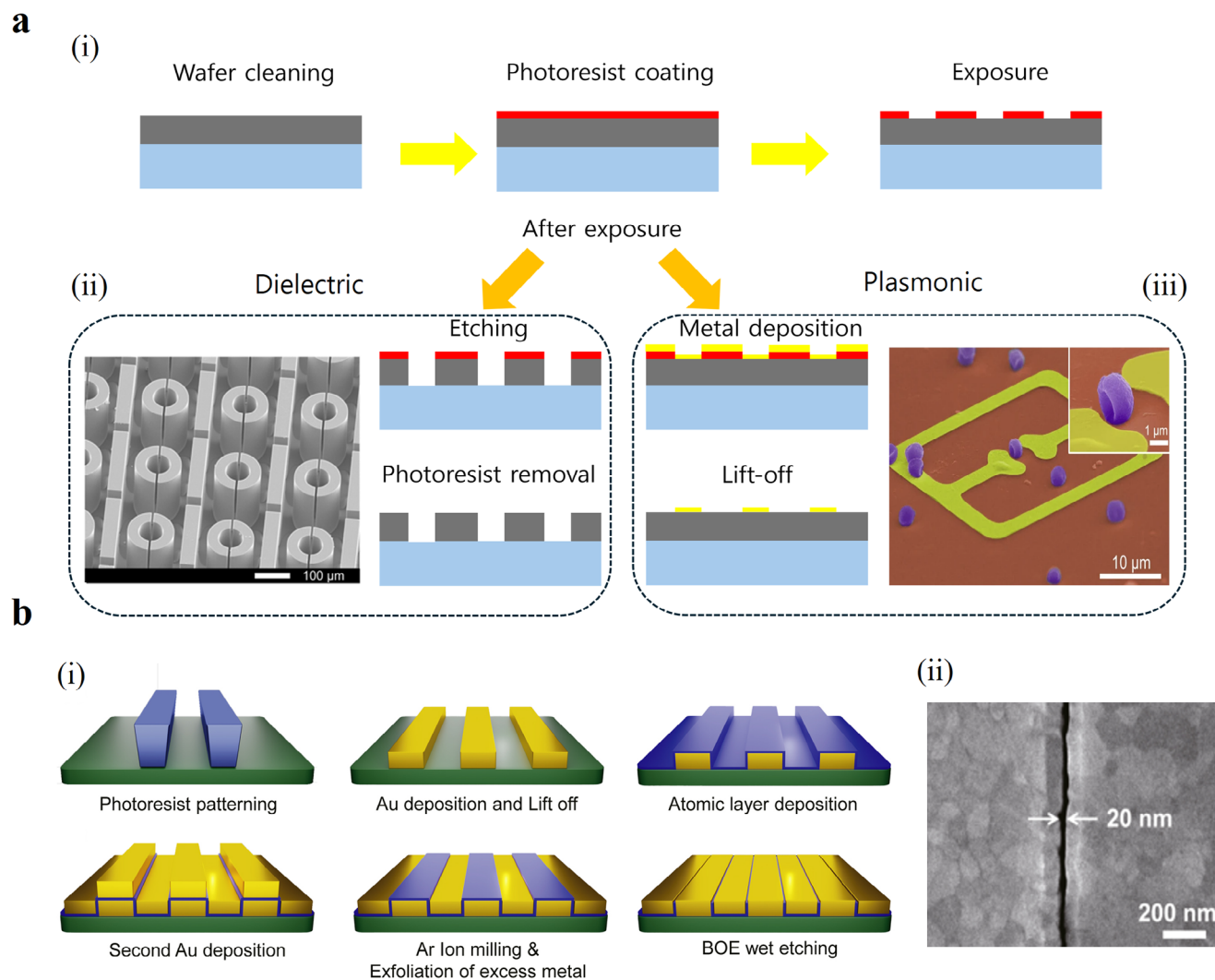


Figure 8. a-i) Schematic of the photolithography process. ii) After exposure, the dielectric metasensor is created using patterned PR as an etch mask. iii) After exposure, a metal film is deposited, and the PR is subsequently removed through a lift-off process, leaving behind the metal nanostructures. b-i) Schematic of atomic layer lithography fabrication method. ii) SEM image of fabricated nanogap structures with a gap size of 20 nm. a-ii) Reproduced with permission.^[191] Copyright 2024, MDPI. a-iii) Reproduced with permission.^[192] Copyright 2014, Springer Nature. b) Reproduced with permission.^[181] Copyright 2023, De Gruyter.

characteristics. To address this challenge, strategies generally fall into two categories: environment-focused approaches that aim to reduce water vapor absorption along the THz beam path, and sample-specific approaches that minimize interaction between THz waves and water within the analyte itself. The first category targets environmental moisture in the free-space propagation path of THz waves. For example, tightly integrating the emitter, sample region, and detector into a compact system reduces the THz beam's exposure to humid air, thereby minimizing background absorption. Additionally, purging the optical path or measurement chamber with dry air or nitrogen gas is commonly used to displace ambient water vapor and stabilize the baseline signal.^[182] In contrast, the second category focuses on mitigating water absorption within aqueous samples, which is especially important for biosensing applications. One effective strategy is

the use of microfluidic platforms with thin flow channels, which limit the optical path length through liquid water while maintaining biological activity. Another approach involves nanogap confinement, wherein analyte molecules are confined within high-field nanostructures that enhance local interactions while reducing the overall water background.^[183] Interestingly, studies have shown that water confined in such nanogaps can even exhibit ice-like behavior, offering unique sensing opportunities.^[146] Furthermore, attenuated total reflection (ATR) geometries utilize evanescent fields to probe surface-bound species, thereby avoiding direct THz transmission through the liquid bulk and substantially reducing water-induced losses.^[184] Together, these strategies expand the capabilities of THz metasensors in aqueous and bio-relevant environments, allowing for high-S detection in both ambient and liquid-phase conditions.

5. Fabrication

Over the past few decades, there have been remarkable advancements in nanomanufacturing technology, such as focused ion beam milling,^[185] electron beam lithography,^[186–187] photolithography,^[188] and nanoimprint lithography,^[189–190] which have significantly enhanced the precision and capabilities of nanopatterning techniques. This section provides an overview of common fabrication methods specifically used for THz metasensors, which can be categorized into mask-based and mask-free approaches. Mask-based methods involve the use of masks to define patterns on the substrate, typically requiring multiple steps for pattern transfer. In contrast, mask-free methods eliminate the need for physical masks, offering more flexibility and often simplifying the fabrication process by directly writing or assembling structures onto the substrate. These approaches will be discussed in separate sections.

5.1. Mask-Based Methods

In this section, we will discuss mask-based methods, highlighting the most widely used techniques for large-scale production, such as photolithography and atomic layer lithography. Mask-based methods offer several advantages, including high precision and reproducibility, which are essential for producing intricate patterns with high accuracy. These techniques also enable scalability, making them ideal for the mass production of THz metasensors and other nanostructures.

5.1.1. Photolithography

Currently, most THz metasensor fabrication methods primarily rely on the photolithography technique. **Figure 8a(i)** schematically illustrates the photolithography process, which begins with substrate cleaning, followed by coating them with a photoresist (PR) layer. PR can be classified into positive and negative types, where positive PR becomes soluble in the developer after exposure, and negative PR becomes insoluble. The choice of which type of PR to use can be made by the user, depending on the specific fabrication method and the requirements of the process. A pre-baking step is then followed to remove the solvent from the PR. The coated substrates are then exposed to UV light using a photolithography system. After exposure, the developed area is washed away with a developer solution, leaving the PR with the desired structural pattern as a mask. The subsequent processes differ depending on whether the metasensor to be fabricated is dielectric or metal. In the case of dielectric structures, the PR pattern is used as an etch mask to etch the underlying layer.^[193] However, for THz dielectric metasurfaces, the dimensions of the meta-atoms typically range from tens to hundreds of micrometers, while the thickness of these meta-atoms, created through dry etching, is significantly larger than the meta-atom size. This results in a much higher aspect ratio compared to the nanoscale thickness of optical dielectric metasurfaces made with inductively coupled plasma techniques. Therefore, deep reactive ion etching, particularly the Bosch process, is commonly used to etch these structures, to achieve such high aspect ratios.^[194]

Men et al. (2024) demonstrated an all-dielectric metamaterial THz biosensor with a high Q-factor of 35 at a 0.82 THz resonance peak, offering simpler fabrication and enhanced EM confinement. The microstructure and unit cell dimensions of the device are shown in **Figure 8a(ii)**.^[191] In the case of plasmonic structure, a thin metal film was deposited onto the coated substrates using magnetron sputtering or e-beam evaporation. To enhance the adhesion of the metal film to the substrate, a thin layer of Cr or Ti, typically a few nanometers thick, is often pre-deposited before the metal deposition. Afterward, a lift-off process was performed to remove the PR, leaving behind the complete metamaterial structure. Park et al. (2014) demonstrated THz metamaterials for the rapid and highly sensitive detection of microorganisms using photolithography. The metamaterial patterns consisted of split ring resonators with a linewidth of 4 μm and gap sizes of 2–3 μm . By leveraging the resonance frequency shift of metamaterials, which is affected by the number density and dielectric constants of micro-organisms (**Figure 8a(iii)**).^[192]

5.1.2. Atomic Layer Lithography

Nanogap resonators have been proposed to enhance the S of THz sensing.^[11] These resonators increase capacitance and enhance field confinement around the nanogap, resulting in improved S. To fabricate these structures, a technique known as atomic layer lithography has been introduced.^[195] **Figure 8b(i)**^[183] illustrates the following fabrication method. Ji et al. (2023) demonstrated virus sensing via nanogap resonators using atomic layer lithography. The n-type PR was spin-coated, baked, and exposed to UV light before being developed. Cr and Au layers were deposited on the patterns using electron-beam evaporation, followed by a lift-off process. An insulating Al_2O_3 layer was deposited via atomic layer deposition, setting the nanogap width, and the gaps are filled with gold through secondary Au deposition. Excess gold was removed by exfoliation using a tape, and the remaining Al_2O_3 layer on the gold surface was etched using a buffered oxide etchant. The fabricated structure consists of 20 nm gap rectangular loop patterns that resonate at THz frequencies, providing a highly responsive platform for detecting virus particles (**Figure 8b(ii)**).^[183]

5.2. Mask-Less Methods

In this section, we will discuss mask-less methods such as direct writing, μSL , TPP printing, and self-assembly, in addition to traditional photolithography, etching techniques, and the more complex atomic layer lithography method mentioned earlier. Mask-less methods offer the advantage of simplified fabrication processes, eliminating the need for masks and enabling more flexible, cost-effective production. We will now explore the advantages and disadvantages of each of these techniques, along with the structures that have been created using these methods.

5.2.1. Self-Assembly

Self-assembly is a fabrication method where materials such as molecules or nanoparticles spontaneously organize into

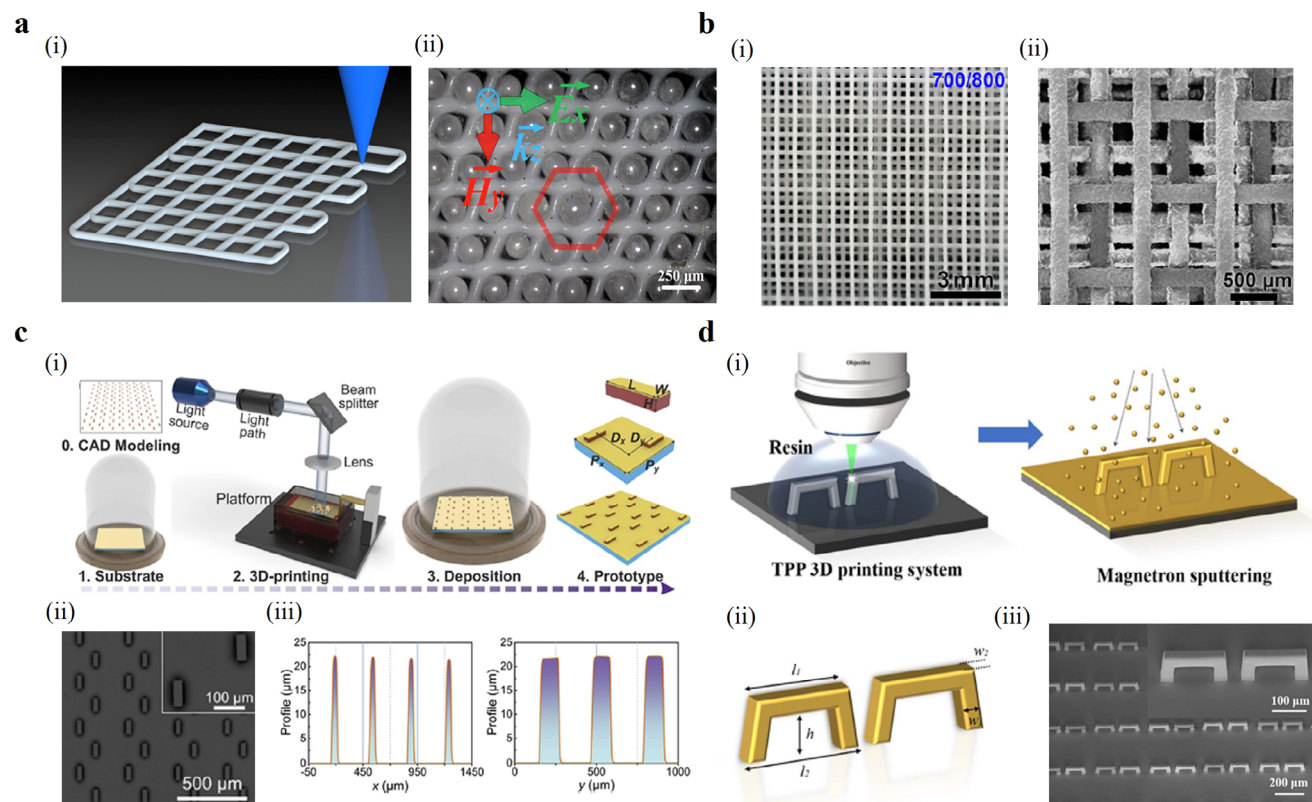


Figure 9. a-i) Schematic of the 2D template using the 3D direct writing technique. ii) ZrO_2 microspheres with a diameter of $130\ \mu\text{m}$ placed on a PDMS template. b-i) Optical microscope image of 3D THz photonic crystal using direct ink writing. ii) Scanning electron microscope (SEM) image of the same structure. c-i) Schematic of Projection μSL fabrication method. ii) SEM image of fabricated structures with periodic bar-like meta-atoms. iii) Surface profiles along the x and y axes of the structure. d-i) Fabrication process of THz meta-absorber. ii) Geometry of the meta-atom iii) SEM images of the fabricated metasurface. a) Reproduced with permission.^[198] Copyright 2018, Optical Society of America. b) Reproduced with permission.^[200] Copyright 2023, Elsevier. c) Reproduced with permission.^[202] Copyright 2022, Optical Society of America. d) Reproduced with permission.^[97] Copyright 2024, Optical Society of America.

structured patterns, typically driven by specific interactions like van der Waals forces and electrostatic forces. It is commonly used to fabricate micro- and nanoscale structures, often guided by templates. Self-assembly is more cost-effective and simpler than traditional photolithography and etching methods, making it suitable for large-scale manufacturing.^[196] Furthermore, since self-assembly relies on templates, it is more suitable for creating simple periodic structures. Self-assembly has also emerged as an effective approach for creating THz metasurfaces using high-dielectric constant microscale spheres.^[197] By manipulating the conditions of self-assembly, these microspheres can organize into either ordered or disordered structures, achieving the desired optical response. Researchers have developed a self-assembly method combined with 2D templates (Figure 9a(i)).^[198] Gao et al. (2018) produced ZrO_2 microspheres with a diameter of $130\ \mu\text{m}$ using the reaction spray atomization method. They placed the microspheres on a PDMS grid through 3D layer-by-layer printing (Figure 9a(ii)).^[198] A $40\ \mu\text{m}$ -thick Kapton adhesive layer was used to secure the microspheres to the grid. Any excess microspheres were easily removed without disturbing the adhesive layer. In this setup, the PDMS grid acted as support for the microspheres.

5.2.2. Direct Ink Writing

Direct ink writing is an additive manufacturing technique that uses a computer-controlled system to deposit material layer by layer, enabling the creation of complex 3D structures.^[199] Using this method, flexible 3D THz photonic crystals can be fabricated with a composite ink of PDMS and barium titanate (BaTiO_3) nanoparticles, allowing for precise control over geometry and THz properties by adjusting the nanoparticle content. Using this method, Zhu et al. (2017) developed a composite ink containing barium titanate (BaTiO_3) nanoparticles dispersed in PDMS. They utilized this ink to print flexible 3D THz photonic crystals with periodically cross-stacked structures, layer by layer, as depicted in Figure 9b(i,ii).^[200] This approach enables the precise fabrication of complex 3D structures, offering enhanced flexibility and S for THz applications.

5.2.3. Projection Micro Stereolithography

Projection μSL 3D printing offers a simpler, yet highly effective procedure compared to traditional micro-nano fabrication techniques,^[201] making it a practical option for creating

out-of-plane micro-structures. The process begins with creating a 3D model, which is sliced into 2D black-and-white digital images. These 2D images are subsequently transferred to the 3D printer, where they are used to control the UV light patterns via the digital micromirror device. The shaped UV light then passes through a projection lens, which projects the 2D images onto the resin surface, initiating the localized photopolymerization in the patterned areas to form a layer of the printed structure. After printing one layer, the translational stage controller then lowers the sample, and the UV projection continues to photopolymerize the next layer. This process repeats iteratively until the entire structure is completed. Once the structure is 3D printed, a metal layer is deposited onto the printed structure using electron beam evaporation (Figure 9c(ii)).^[202] Using this method, Yin et al. (2022) recently developed a highly sensitive THz sensor with an out-of-plane metasurface consisting of periodically arranged bar-like meta-atoms, as shown in Figure 9c(ii).^[202] The prototype of the proposed THz sensor is shown schematically, featuring an array of meta-molecules with dimensions of $L = 105 \mu\text{m}$, $W = 35 \mu\text{m}$, and $H = 20 \mu\text{m}$, arranged in a 30×30 grid (Figure 9c(iii)).^[202] This sensor demonstrates a high S of 325 GHz RIU^{-1} , attributed to toroidal dipole resonance-enhanced THz-matter interactions.

5.2.4. Two-Photon Polymerization 3D Printing

TPP 3D printing is a fabrication method using a femtosecond laser direct writing system, enabling the creation of diverse, high-precision 3D metasurfaces, thus offering a new approach for designing multi-functional 3D metasurfaces.^[203] Using this method, Chen et al. (2024) recently developed a 3D meta-absorber with an out-of-plane plasma mechanism based on two-photon printing, offering simpler fabrication and enhanced EM confinement as shown in the Figure 9d.^[97] The fabrication process is as follows: The structure is printed by selectively curing the resin with a laser, with real-time monitoring during the process. After printing, uncured resin is removed, and UV light curing is applied to enhance structural strength, addressing adhesion and stability issues through methods like plasma pre-treatment and proper printing dose (Figure 9d(i)).^[97] The design features a gate-frame structure with copper-coated resin, with geometric parameters including $p = 400 \mu\text{m}$, $l_1 = 140 \mu\text{m}$, $l_2 = 160 \mu\text{m}$, $h = 80 \mu\text{m}$, $w_1 = 25 \mu\text{m}$, and $w_2 = 30 \mu\text{m}$. The copper thickness is set as 500 nm , ensuring reduced THz transmission (Figures 9d(ii,iii)).^[97] This structure demonstrated high- S for doxycycline detection in the THz spectrum, with potential applications in trace antibiotic sensing. TPP offers several advantages, including high resolution and precision as low as 100 nm , which allows for the creation of nanometer-scale structures. The nonlinear two-photon absorption enables accuracy beyond the diffraction limit, and the technique provides spatial flexibility and precise control for fabricating complex 3D structures. However, TPP has limitations, such as slow production speed, which makes it less efficient for large-scale manufacturing.^[204]

6. Conclusion and Outlook

In this review, we have provided a comprehensive overview of the current state of THz metasensors, encompassing their un-

derlying physical mechanisms, key performance metrics, diverse sensing applications, and fabrication strategies. The unique capability of THz metasurfaces to tailor light-matter interactions at the subwavelength scale has enabled breakthroughs in label-free and highly sensitive detection of biological, chemical, and physical phenomena. Particularly, metasensors leveraging resonance- and interference-based mechanisms have demonstrated significant potential for real-time detection and identification of molecular fingerprints, offering new paradigms for biomedical diagnostics, environmental monitoring, and material analysis.

The weak interaction between THz waves and nanometer-scale analytes has traditionally limited sensing performance under ambient conditions. Recent advances in nanogap-engineered metamaterials have significantly alleviated this challenge by providing strong local field enhancement and efficient light-matter coupling at subwavelength scales. Despite these improvements, several key challenges remain. These include achieving high- Q resonances with wide tunability, ensuring thermal and mechanical stability, and maintaining compatibility with large-area, low-cost manufacturing. Integration with portable electronics and real-time readout systems is also essential for developing field-deployable THz sensing platforms. Although metasensors offer a pathway to compact and low-cost chip designs, the overall system cost is still high. This is primarily due to the lack of affordable and integrated THz emitters and detectors. Current THz sources rely on bulky and expensive components, such as femtosecond lasers and photoconductive antennas, while compact alternatives remain in early development. Progress in low-cost, integrable THz transceivers must accompany advances in sensor design to enable widespread deployment.

Future progress will likely be driven by interdisciplinary approaches that combine materials innovation, non-Hermitian physics, and data-driven design. For instance, the introduction of exceptional points and topological effects in non-Hermitian metasurfaces has opened new possibilities for enhanced S and spectral resolution. Additionally, the convergence of artificial intelligence with inverse design is accelerating the discovery of optimal metasurface configurations tailored for specific analytes and environments. The use of active materials such as graphene, conductive polymers, and phase-change media offers promising routes toward tunable and multifunctional sensing devices.

On the fabrication front, various fabrication strategies have been covered in this review, ranging from conventional photolithography to advanced techniques such as atomic layer lithography, self-assembly, TPP, etc. These methods have been developed to construct complex nanostructures with high fidelity and functional versatility. Progress in hybrid and roll-to-roll fabrication approaches is also expected to reduce production costs while maintaining high performance and reproducibility.

Looking ahead, THz metasensors are poised to become key components of next-generation detection platforms, not only for high-resolution laboratory spectroscopy but also for compact, in-situ, and real-time sensing across diverse domains. As the field continues to evolve through synergistic developments in photonics, materials science, and artificial intelligence, we anticipate that THz metasensors will serve as a cornerstone technology for addressing urgent needs in healthcare, environmental monitoring, security, and beyond.

Acknowledgements

This work was financially supported by the POSCO-POSTECH-RIST Convergence Research Center program funded by POSCO, and the National Research Foundation (NRF) (Grant No. RS-2025-25396100) funded by the Ministry of Science and ICT (MSIT) of the Korean government. X.L. acknowledges the China Scholarship Council (202306890039). D.H. acknowledges the NRF Quantum Information Science fellowship (RS-2024-00402302) funded by the MSIT of the Korean government. The authors thank Prof. Xianzhong Chen (Heriot-Watt University) for discussions.

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

X.L., H.-T.L., and D.H. contributed equally to this work. J.R. conceived the idea and initiated the project. X.L., H.-T.L., and D. H. mainly wrote the manuscript. All authors confirmed the final manuscript. J.R. guided the entire work.

Keywords

chiral sensing, Fano resonances, metasensor, metasurface, quasi-bound state in the continuum, terahertz sensing

Received: April 27, 2025

Revised: June 30, 2025

Published online: July 18, 2025

- [1] D. L. Woolard, M. Shur, W. R. Loerop, World Scientific (Firm), Eds., *Terahertz Sensing Technology: Vol. 1: Electronic Devices and Advanced Systems Technology*, World Scientific Pub. Co, Singapore River Edge, NJ **2003**.
- [2] B. Ferguson, X.-C. Zhang, *Nat. Mater.* **2002**, *1*, 26.
- [3] F. Sizov, A. Rogalski, *Prog. Quantum Electron.* **2010**, *34*, 278.
- [4] D. L. Woolard, R. Brown, M. Pepper, M. Kemp, *Proc. IEEE* **2005**, *93*, 1722.
- [5] T. Jia, B. Li, Z. Zhang, B. Gu, G. Rui, *Results Phys.* **2023**, *52*, 106857.
- [6] T. Jia, B. Li, B. Gu, Q. Zhan, G. Rui, *Opt. Laser Technol.* **2025**, *181*, 111995.
- [7] A. Ahmadvand, B. Gerislioglu, R. Ahuja, Y. K. Mishra, *Mater. Today* **2020**, *32*, 108.
- [8] A. Ahmadvand, B. Gerislioglu, *Laser Photonics Rev.* **2022**, *16*, 2100328.
- [9] Y. Kim, H. Kim, Y. Yang, T. Badloe, N. Jeon, J. Rho, *Nanoscale* **2022**, *14*, 3720.
- [10] S. Shen, X. Liu, Y. Shen, J. Qu, E. Pickwell-MacPherson, X. Wei, Y. Sun, *Adv. Opt. Mater.* **2022**, *10*, 2101008.
- [11] Y.-M. Bahk, K.-H. Kim, K. J. Ahn, H.-R. Park, *Adv. Photonics Res.* **2024**, *5*, 2300211.
- [12] J. Lyu, L. Huang, L. Chen, Y. Zhu, S. Zhuang, *Photonics Res.* **2024**, *12*, 194.
- [13] X. Chen, S. Xin, Q. Liu, Y. Meng, D. Yu, M. L. Tseng, L. Ye, *Nanophotonics* **2025**, *14*, 271.
- [14] H.-H. Hsiao, C. H. Chu, D. P. Tsai, *Small Methods* **2017**, *1*, 1600064.
- [15] M. Khorasaninejad, F. Capasso, *Science* **2017**, *358*, aam8100.
- [16] Q. He, S. Sun, S. Xiao, L. Zhou, *Adv. Opt. Mater.* **2018**, *6*, 1800415.
- [17] L. Huang, S. Zhang, T. Zentgraf, *Nanophotonics* **2018**, *7*, 1169.
- [18] S. Chen, W. Liu, Z. Li, H. Cheng, J. Tian, *Adv. Mater.* **2020**, *32*, 1805912.
- [19] W. T. Chen, A. Y. Zhu, F. Capasso, *Nat. Rev. Mater.* **2020**, *5*, 604.
- [20] R. Zhao, L. Huang, Y. Wang, *Photonix* **2020**, *1*, 20.
- [21] C.-W. Qiu, T. Zhang, G. Hu, Y. Kivshar, *Nano Lett.* **2021**, *21*, 5461.
- [22] C. Jung, G. Kim, M. Jeong, J. Jang, Z. Dong, T. Badloe, J. K. W. Yang, J. Rho, *Chem. Rev.* **2021**, *121*, 13013.
- [23] N. Wang, W. Yan, Y. Qu, S. Ma, S. Z. Li, M. Qiu, *Photonix* **2021**, *2*, 22.
- [24] J. Kim, J. Seong, Y. Yang, S.-W. Moon, T. Badloe, J. Rho, *Adv. Photonics* **2022**, *4*, 024001.
- [25] J. Qin, S. Jiang, Z. Wang, X. Cheng, B. Li, Y. Shi, D. P. Tsai, A. Q. Liu, W. Huang, W. Zhu, *ACS Nano* **2022**, *16*, 11598.
- [26] Q. Song, X. Liu, C.-W. Qiu, P. Genevet, *Appl. Phys. Rev.* **2022**, *9*, 011311.
- [27] S. So, J. Mun, J. Park, J. Rho, *Adv. Mater.* **2023**, *35*, 2206399.
- [28] H. Yang, K. Ou, H. Wan, Y. Hu, Z. Wei, H. Jia, X. Cheng, N. Liu, H. Duan, *Mater. Today* **2023**, *67*, 424.
- [29] T. J. Cui, S. Zhang, A. Alù, M. Wegener, S. J. Pendry, J. Luo, Y. Lai, Z. Wang, X. Lin, H. Chen, P. Chen, R.-X. Wu, Y. Yin, P. Zhao, H. Chen, Y. Li, Z. Zhou, N. Engheta, V. Asadchy, C. Simovski, S. Tretyakov, B. Yang, S. D. Campbell, Y. Hao, D. H. Werner, S. Sun, L. Zhou, S. Xu, H.-B. Sun, Z. Zhou, et al., *J. Phys. Photonics* **2024**, *6*, 032502.
- [30] X. Hu, W. Xu, Q. Fan, T. Yue, F. Yan, Y. Lu, T. Xu, *Adv. Photonics* **2024**, *6*, 014002.
- [31] Y. Kim, X. Li, P. Tang, G. Li, J. Rho, *ACS Photonics* **2025**, *12*, 16.
- [32] P. Tang, Y. Jeon, X. Li, Y. Kim, J. Seong, S. Hu, K. Kim, G. Li, T. Badloe, J. Rho, *Adv. Opt. Mater.* **2025**, 2500981.
- [33] N. M. Estakhri, A. Alù, *Phys. Rev. X* **2016**, *6*, 041008.
- [34] A. Komar, R. Paniagua-Domínguez, A. Miroshnichenko, Y. F. Yu, Y. S. Kivshar, A. I. Kuznetsov, D. Neshev, *ACS Photonics* **2018**, *5*, 1742.
- [35] S.-Q. Li, X. Xu, R. Maruthiyodan Veetil, V. Valuckas, R. Paniagua-Domínguez, A. I. Kuznetsov, *Science* **2019**, *364*, 1087.
- [36] X. Fu, F. Yang, C. Liu, X. Wu, T. J. Cui, *Adv. Opt. Mater.* **2020**, *8*, 1900628.
- [37] J. Wen, L. Chen, X. Chen, S. Kanwal, L. Zhang, S. Zhuang, D. Zhang, D. Lei, *Laser Photonics Rev.* **2021**, *15*, 2000487.
- [38] X. Cai, R. Tang, H. Zhou, Q. Li, S. Ma, D. Wang, T. Liu, X. Ling, W. Tan, Q. He, S. Xiao, L. Zhou, *Adv. Photonics* **2021**, *3*, 036003.
- [39] Q. Li, X. Cai, T. Liu, M. Jia, Q. Wu, H. Zhou, H. Liu, Q. Wang, X. Ling, C. Chen, F. Ding, Q. He, Y. Zhang, S. Xiao, L. Zhou, *Nanophotonics* **2022**, *11*, 2085.
- [40] X. Ni, A. V. Kildishev, V. M. Shalae, *Nat. Commun.* **2013**, *4*, 2807.
- [41] L. Li, T. Jun Cui, W. Ji, S. Liu, J. Ding, X. Wan, Y. Bo Li, M. Jiang, C.-W. Qiu, S. Zhang, *Nat. Commun.* **2017**, *8*, 197.
- [42] Y. Hu, L. Li, Y. Wang, M. Meng, L. Jin, X. Luo, Y. Chen, X. Li, S. Xiao, H. Wang, Y. Luo, C.-W. Qiu, H. Duan, *Nano Lett.* **2020**, *20*, 994.
- [43] P. Georgi, Q. Wei, B. Sain, C. Schlickriede, Y. Wang, L. Huang, T. Zentgraf, *Sci. Adv.* **2021**, *7*, abf9718.
- [44] I. Kim, W.-S. Kim, K. Kim, M. A. Ansari, M. Q. Mehmood, T. Badloe, Y. Kim, J. Gwak, H. Lee, Y.-K. Kim, J. Rho, *Sci. Adv.* **2021**, *7*, abe9943.
- [45] A. Asad, J. Seong, H. S. Khaliq, J. Lee, Y. Jeon, H. Cabrera, M. Q. Mehmood, L. Gao, H. Kim, J. Rho, *Adv. Funct. Mater.* **2024**, *34*, 2402744.
- [46] J. Ko, G. Kim, I. Kim, S. H. Hwang, S. Jeon, J. Ahn, Y. Jeong, J. Ha, H. Heo, J. Jeong, I. Park, J. Rho, *Adv. Sci.* **2024**, *11*, 2407045.
- [47] K. Chen, Y. Feng, F. Monticone, J. Zhao, B. Zhu, T. Jiang, L. Zhang, Y. Kim, X. Ding, S. Zhang, A. Alù, C. Qiu, *Adv. Mater.* **2017**, *29*, 1606422.
- [48] W. T. Chen, A. Y. Zhu, V. Sanjeev, M. Khorasaninejad, Z. Shi, E. Lee, F. Capasso, *Nat. Nanotechnol.* **2018**, *13*, 220.
- [49] R. Paniagua-Domínguez, Y. F. Yu, E. Khaidarov, S. Choi, V. Leong, R. M. Bakker, X. Liang, Y. H. Fu, V. Valuckas, L. A. Krivitsky, A. I. Kuznetsov, *Nano Lett.* **2018**, *18*, 2124.

- [50] R. J. Lin, V.-C. Su, S. Wang, M. K. Chen, T. L. Chung, Y. H. Chen, H. Y. Kuo, J.-W. Chen, J. Chen, Y.-T. Huang, J.-H. Wang, C. H. Chu, P. C. Wu, T. Li, Z. Wang, S. Zhu, D. P. Tsai, *Nat. Nanotechnol.* **2019**, *14*, 227.
- [51] F. Balli, M. Sultan, S. K. Lami, J. T. Hastings, *Nat. Commun.* **2020**, *11*, 3892.
- [52] T. Badloe, Y. Kim, J. Kim, H. Park, A. Barulin, Y. N. Diep, H. Cho, W.-S. Kim, Y.-K. Kim, I. Kim, J. Rho, *ACS Nano* **2023**, *17*, 14678.
- [53] T. Badloe, J. Seong, J. Rho, *Nano Lett.* **2023**, *23*, 6958.
- [54] Z. Huang, Y. Zheng, J. Li, Y. Cheng, J. Wang, Z.-K. Zhou, L. Chen, *Nano Lett.* **2023**, *23*, 10991.
- [55] F. Yang, M. Y. Shalaginov, H.-I. Lin, S. An, A. Agarwal, H. Zhang, C. Rivero-Baleine, T. Gu, J. Hu, *Adv. Photonics* **2023**, *5*, 033001.
- [56] X. Li, X. Cai, C. Liu, Y. Kim, T. Badloe, H. Liu, J. Rho, S. Xiao, *Opto-Electron. Adv.* **2024**, *7*, 240085.
- [57] M. Choi, J. Kim, S. Moon, K. Shin, S.-W. Nam, Y. Park, D. Kang, G. Jeon, K. Lee, D. H. Yoon, Y. Jeong, C.-K. Lee, J. Rho, *Nat. Mater.* **2025**, *24*, 535.
- [58] J. Kim, J. An, W. Kim, J. Seong, Y. Park, E. Lee, S. Kim, S. Moon, C. Lee, H. Lee, J. Rho, *Laser Photonics Rev.* **2025**, *19*, 2401425.
- [59] Y. Kim, J. Lee, W.-H. Yeo, X. Li, W.-S. Kim, Y.-K. Kim, T. Badloe, C. Sun, H. F. Zhang, J. Rho, *Nano Lett.* **2025**, *25*, 9394.
- [60] X. Yin, Z. Ye, J. Rho, Y. Wang, X. Zhang, *Science* **2013**, *339*, 1405.
- [61] A. Shaltout, J. Liu, A. Kildishev, V. Shalae, *Optica* **2015**, *2*, 860.
- [62] S. Xiao, F. Zhong, H. Liu, S. Zhu, J. Li, *Nat. Commun.* **2015**, *6*, 8360.
- [63] W. Zhu, R. Yang, G. Geng, Y. Fan, X. Guo, P. Li, Q. Fu, F. Zhang, C. Gu, J. Li, *Nanophotonics* **2020**, *9*, 4327.
- [64] M. Kim, D. Lee, Y. Yang, Y. Kim, J. Rho, *Nat. Commun.* **2022**, *13*, 2036.
- [65] Y. Ke, Y. Bian, Q. Tang, J. Tian, L. Zeng, Y. Chen, X. Zhou, *Nanophotonics* **2023**, *12*, 4361.
- [66] P. Tang, X. Li, Y. Kim, L. Xiao, H. Wu, T. Badloe, G. Li, J. Rho, *ACS Photonics* **2024**, *11*, 4170.
- [67] J. Lee, J. Kim, S. Shim, Y. Yang, J. Choi, J. Rho, D. Lee, M. Kim, *Nat. Commun.* **2025**, *16*, 2699.
- [68] Y. Kim, P. Tang, C. Jung, J. Kim, J. Lee, X. Li, H. Keawmuang, S. Hu, G. Li, T. Badloe, J. Rho, *Laser Photonics Rev.* **2025**, 00581.
- [69] S. Chen, Y. Cai, G. Li, S. Zhang, K. W. Cheah, *Laser Photonics Rev.* **2016**, *10*, 322.
- [70] F. Yue, D. Wen, J. Xin, B. D. Gerardot, J. Li, X. Chen, *ACS Photonics* **2016**, *3*, 1558.
- [71] H.-X. Xu, H. Liu, X. Ling, Y. Sun, F. Yuan, *IEEE Trans. Antennas Propag.* **2017**, *65*, 7378.
- [72] F. Ding, Y. Chen, S. I. Bozhevolnyi, *Nanophotonics* **2020**, *9*, 371.
- [73] Q. Zhou, M. Liu, W. Zhu, L. Chen, Y. Ren, H. J. Lezec, Y. Lu, A. Agrawal, T. Xu, *Laser Photonics Rev.* **2021**, *15*, 2100390.
- [74] H. Ahmed, H. Kim, Y. Zhang, Y. Intaravanne, J. Jang, J. Rho, S. Chen, X. Chen, *Nanophotonics* **2022**, *11*, 941.
- [75] K. A. Marx, *Biomacromolecules* **2003**, *4*, 1099.
- [76] J. Homola, *Chem. Rev.* **2008**, *108*, 462.
- [77] R. M. Lequin, *Clin. Chem.* **2005**, *51*, 2415.
- [78] M. M. Hassan, F. S. Sium, F. Islam, S. M. Choudhury, *Sens. Bio-Sens. Res.* **2021**, *33*, 100429.
- [79] J. Zhu, C. Li, B. Fang, J. Lu, Z. Hong, X. Jing, *Opt. Laser Technol.* **2024**, *176*, 110978.
- [80] Z. Zhang, R. Zhao, M. Cong, J. Qiu, *Nanotechnol. Rev.* **2024**, *13*, 20230182.
- [81] Y. Wang, D. Zhu, Z. Cui, L. Hou, L. Lin, F. Qu, X. Liu, P. Nie, *ACS Omega* **2019**, *4*, 18645.
- [82] Y. Huang, S. Zhong, T. Shi, Y. Shen, D. Cui, *Nanophotonics* **2020**, *9*, 3011.
- [83] Y. Huang, R. Yuan, S. Zhong, T. Shi, F. Sun, T. Lin, Y. Zhong, *IEEE Sens. J.* **2024**, *24*, 27455.
- [84] W. Yu, J. Li, G. Huang, Z. He, H. Tian, F. Xie, W. Jin, Q. Huang, W. Fu, X. Yang, *Talanta* **2024**, *272*, 125760.
- [85] T. Mu, Y. Ye, Z. Dai, R. Zhao, M. Yang, X. Ren, *Opt. Express* **2022**, *30*, 41101.
- [86] W. Wang, M. Sun, J. Lin, Y. Xue, Y. Shi, *Biosensors* **2024**, *14*, 318.
- [87] W. Shi, F. Fan, S. Li, L. Ma, Y.-M. Wang, X. Wang, S. Chang, *IEEE Sens. J.* **2024**, *24*, 7690.
- [88] S. J. Park, S. H. Cha, G. A. Shin, Y. H. Ahn, *Biomed. Opt. Express* **2017**, *8*, 3551.
- [89] W. Shi, F. Fan, S. Li, Z. Zhang, H. Liu, X. Wang, S. Chang, *Sens. Actuators B: Chem.* **2022**, *362*, 131777.
- [90] X. Lu, F. Zhang, L. Zhu, S. Peng, J. Yan, Q. Shi, K. Chen, X. Chang, H. Zhu, C. Zhang, W. Huang, Q. Cheng, *Nat. Commun.* **2024**, *15*, 3157.
- [91] X. Zhang, Y. Wang, Y. Ru, X. Zhang, F. Luo, X. Chen, X. Zhao, C. Wang, *Talanta* **2025**, *286*, 127549.
- [92] A. Ahmadivand, B. Gerislioglu, Z. Ramezani, A. Kaushik, P. Manickam, S. A. Ghoreishi, *Biosens. Bioelectron.* **2021**, *117*, 112971.
- [93] J. Xu, D. Liao, M. Gupta, Y. Zhu, S. Zhuang, R. Singh, L. Chen, *Adv. Opt. Mater.* **2021**, *9*, 2100024.
- [94] W. Han, P. Wang, C. Zhang, Y. Xu, Z. Wu, Q. Wang, *ACS Appl. Mater. Interfaces* **2024**, *16*, 70795.
- [95] X. Hu, G. Xu, L. Wen, H. Wang, Y. Zhao, Y. Zhang, D. R. S. Cumming, Q. Chen, *Laser Photonics Rev.* **2016**, *10*, 962.
- [96] X. Chen, W. Fan, C. Song, *Carbon* **2018**, *133*, 416.
- [97] X. Chen, L. Ye, D. Yu, *Photonics Res.* **2024**, *12*, 895.
- [98] M. Manjappa, Y. K. Srivastava, A. Solanki, A. Kumar, T. C. Sum, R. Singh, *Adv. Mater.* **2017**, *29*, 1605881.
- [99] K. Shih, P. Pitchappa, L. Jin, C.-H. Chen, R. Singh, C. Lee, *Appl. Phys. Lett.* **2018**, *113*, 071105.
- [100] K. Serita, H. Murakami, I. Kawayama, M. Tonouchi, *Photonics* **2019**, *6*, 12.
- [101] L. Huang, H. Cao, L. Chen, Y. Ma, Y. Yang, X. Liu, W. Wang, Y. Zhu, S. Zhuang, *Talanta* **2024**, *269*, 125481.
- [102] B. Liu, Y. Peng, Y. Hao, Y. Zhu, S. Chang, S. Zhuang, *Photonix* **2024**, *5*, 10.
- [103] M. Zhang, L. Jiang, S. Wang, S. Zhang, M. Liu, Y. Zhang, H. Zhang, Z. Tian, *Sci. Rep.* **2025**, *15*, 8585.
- [104] P. Cao, C. Li, Y. Li, Y. Wu, X. Chen, T. Wu, X. Zhang, M. Yuan, Z. Wang, *IEEE Sens. J.* **2022**, *22*, 2110.
- [105] H. Tian, Y. Zhang, L. Liang, X. Yan, H. Yao, Z. Wang, Z. Li, X. Hu, Y. Li, J. Wu, S. Ma, Z. Shi, Q. Wang, Y. Chen, F. Huai, *Spectrochim. Acta. A. Mol. Biomol. Spectrosc.* **2025**, *335*, 125996.
- [106] C. Yang, Z. Shen, Y. Cui, N. Zhang, L. Zhang, R. Yan, X. Chen, *Biosens. Bioelectron.* **2025**, *278*, 117351.
- [107] J. Lyu, S. Shen, L. Chen, Y. Zhu, S. Zhuang, *Photonix* **2023**, *4*, 28.
- [108] C. Luo, L. Chen, *IEEE Sens. J.* **2024**, *24*, 17686.
- [109] X. Liu, W. Chen, Y. Ma, Y. Xie, J. Zhou, L. Zhu, Y. Xu, J. Zhu, *Photonics Res.* **2022**, *10*, 2836.
- [110] R. Wang, L. Xu, L. Huang, X. Zhang, H. Ruan, X. Yang, J. Lou, C. Chang, X. Du, *Small* **2023**, *19*, 2301165.
- [111] R. Wang, R. Hao, D. Li, L. Huang, R. Jiang, X. Zhang, X. Yang, L. Wang, S. Wang, Y. Kivshar, C. Chang, *Adv. Mater.* **2025**, *37*, 2418147.
- [112] F. Aieta, P. Genevet, M. Kats, F. Capasso, *Opt. Express* **2013**, *21*, 31530.
- [113] X. Guo, Z. Zhang, M. Yang, P. Bing, X. Yan, Q. Yang, D. Wei, L. Liu, L. Liang, J. Yao, *Opt. Lasers Eng.* **2021**, *142*, 106566.
- [114] G. Wang, F. Zhu, T. Lang, J. Liu, Z. Hong, J. Qin, *Nanoscale Res. Lett.* **2021**, *16*, 109.
- [115] L. Zhu, H. Li, L. Dong, W. Zhou, M. Rong, X. Zhang, J. Guo, *Opt. Mater. Express* **2021**, *11*, 2109.
- [116] S. Lin, Y. Wang, Z. Peng, Z. Chen, F. Hu, *Talanta* **2022**, *248*, 123628.
- [117] C. Weisenstein, M. Richter, A. K. Wigger, A. Bosserhoff, P. H. Bolívar, *Appl. Phys. Lett.* **2022**, *120*, 053702.

- [118] H. Yao, Z. Sun, X. Yan, M. Yang, L. Liang, G. Ma, J. Gao, T. Li, X. Song, H. Zhang, Q. Yang, X. Hu, Z. Wang, Z. Li, J. Yao, *Nanophotonics* **2022**, 11, 1219.
- [119] X. Huang, W. Ye, J. Ran, Z. Zhou, R. Li, B. Gao, *IEEE Sens. J.* **2023**, 23, 3573.
- [120] W. Liu, X. Zhou, S. Zou, Z. Hu, Y. Shen, M. Cai, D. Lin, J. Zhou, X. Deng, T. Guo, J. Lei, *Nanophotonics* **2023**, 12, 1177.
- [121] H. Yao, Z. Sun, L. Liang, X. Yan, Y. Wang, M. Yang, X. Hu, Z. Wang, Z. Li, M. Wang, C. Huang, Q. Yang, Z. Tian, J. Yao, *Photonics Res.* **2023**, 11, 858.
- [122] Z. Fu, J. Chen, X. Chen, Y. Sun, F. Wang, J. Yang, *Biosensors* **2024**, 14, 338.
- [123] X. Zhang, Y. Wang, F. Luo, X. Zhang, X. Chen, Y. Ru, B. Song, Z. Cui, K. Zhang, *Carbon* **2024**, 227, 119273.
- [124] A. Ahmadvand, B. Gerislioglu, P. Manickam, A. Kaushik, S. Bhansali, M. Nair, N. Pala, *ACS Sens.* **2017**, 2, 1359.
- [125] A. Ahmadvand, B. Gerislioglu, A. Tomitaka, P. Manickam, A. Kaushik, S. Bhansali, M. Nair, N. Pala, *Biomed. Opt. Express* **2018**, 9, 373.
- [126] S. W. Jun, Y. H. Ahn, *Nat. Commun.* **2022**, 13, 3470.
- [127] W. Shi, F. Fan, L. Ma, T.-R. Zhang, J.-Y. Liu, J.-R. Cheng, X. Wang, S. Chang, *Opt. Laser Technol.* **2023**, 162, 109274.
- [128] Y. C. Kim, S. W. Jun, S. J. Park, Y. H. Ahn, *Opt. Express* **2024**, 32, 48915.
- [129] C. Luo, T. T. CaiWei, Z. Fan, L. Chen, R. Singh, Y. Zhu, S. Zhuang, *Sens. Actuators B: Chem.* **2024**, 410, 135628.
- [130] Y. Zhao, Z. Hou, B. Yan, X. Cao, B. Su, M. Lv, H. Cui, C. Zhang, *ACS Appl. Mater. Interfaces* **2024**, 16, 52092.
- [131] N. Zhang, F. Gao, R. Wang, Z. Shen, D. Han, Y. Cui, L. Zhang, C. Chang, C. Qiu, X. Chen, *Adv. Mater.* **2025**, 37, 2411490.
- [132] Y. Ye, Y. Zhang, Y. Zhao, Y. Ren, X. Ren, *Opt. Express* **2021**, 29, 15255.
- [133] X. Du, X. Zhang, Y. Wang, G. Ma, Y. Liu, B. Wang, H. Mao, *Opt. Express* **2021**, 29, 36535.
- [134] Y. Zhang, Y. Xu, H. Liu, B. Sun, *Food Chem.* **2022**, 394, 133467.
- [135] X. Deng, Y. Shen, B. Liu, Z. Song, X. He, Q. Zhang, D. Ling, D. Liu, D. Wei, *Biosensors* **2022**, 12, 408.
- [136] X. Lu, H. Ge, Y. Jiang, Y. Zhang, *Biosensors* **2022**, 12, 471.
- [137] X. Du, H. Mao, Y. Yan, X. Zhang, Y. Wang, Y. Liu, B. Wang, X. Yang, Q. Shi, *IEEE Sens. J.* **2022**, 22, 14065.
- [138] W. Fu, L. Sun, H. Cao, L. Chen, M. Zhou, S. Shen, Y. Zhu, S. Zhuang, *IEEE Sens. J.* **2023**, 23, 12775.
- [139] D. Wang, S. Luo, K.-D. Xu, *ACS Appl. Mater. Interfaces* **2024**, 16, 27969.
- [140] M. Sun, J. Lin, Y. Xue, W. Wang, S. Shi, S. Zhang, Y. Shi, *Nanomaterials* **2024**, 14, 883.
- [141] L. Zhang, X. Kong, S. Wang, W. Zhang, L. Wu, X. Liu, J. Yang, J. Li, F. Qu, *Anal. Chim. Acta* **2025**, 1345, 343734.
- [142] H.-R. Park, S. Namgung, X. Chen, S.-H. Oh, *Faraday Discuss.* **2015**, 178, 195.
- [143] H.-R. Park, X. Chen, N.-C. Nguyen, J. Peraire, S.-H. Oh, *ACS Photonics* **2015**, 2, 417.
- [144] I. Jáuregui-López, P. Rodríguez-Ulibarri, S. A. Kuznetsov, N. A. Nikolaev, M. Beruete, *Sensors* **2018**, 18, 3848.
- [145] R. P. Chaudhary, B. Das, S. I. Oh, D.-S. Kim, *Sci. Rep.* **2019**, 9, 17681.
- [146] H. Yang, G. Ji, M. Choi, S. Park, H. An, H.-T. Lee, J. Jeong, Y. D. Park, K. Kim, N. Park, J. Jeong, D.-S. Kim, H.-R. Park, *Sci. Adv.* **2024**, 10, adm7315.
- [147] M. Seo, J. Kyoung, H. Park, S. Koo, H. Kim, H. Bernien, B. J. Kim, J. H. Choe, Y. H. Ahn, H.-T. Kim, N. Park, Q.-H. Park, K. Ahn, D. Kim, *Nano Lett.* **2010**, 10, 2064.
- [148] H.-R. Park, S. Namgung, X. Chen, N. C. Lindquist, V. Giannini, Y. Francescato, S. A. Maier, S.-H. Oh, *Adv. Opt. Mater.* **2015**, 3, 667.
- [149] J. Jeong, D.-S. Kim, H.-R. Park, *Nanophotonics* **2022**, 11, 3159.
- [150] J. Li, C. M. Shah, W. Withayachumnankul, B. S.-Y. Ung, A. Mitchell, S. Sriram, M. Bhaskaran, S. Chang, D. Abbott, *Opt. Lett.* **2013**, 38, 2104.
- [151] H. J. Shin, S.-H. Kim, K. Park, M.-C. Lim, S.-W. Choi, G. Ok, *Sens. Actuators A: Phys.* **2017**, 268, 27.
- [152] H. S. Kim, S. H. Cha, B. Roy, S. Kim, Y. H. Ahn, *Opt. Express* **2018**, 26, 33575.
- [153] Y. Liu, Y. Chen, J. Li, C. Zhang, Q. Wu, N. Su, M. Li, *Micromachines* **2023**, 14, 816.
- [154] Q. Ma, Y. Ma, Z. Ren, L. Chen, *Opt. Commun.* **2024**, 557, 130362.
- [155] X. Lu, F. Zhang, L. Zhu, S. Peng, J. Yan, Q. Shi, K. Chen, X. Chang, H. Zhu, C. Zhang, *Nat. Commun.* **2024**, 15, 3157.
- [156] W.-H. Lai, B. Li, S.-H. Fu, Y.-S. Lin, *Micromachines* **2023**, 14, 169.
- [157] R. Wang, Q. Wu, Y. Zhang, X. Xu, Q. Zhang, W. Zhao, B. Zhang, W. Cai, J. Yao, J. Xu, *Appl. Phys. Lett.* **2019**, 114, 121102.
- [158] L. Sun, L. Xu, J. Wang, Y. Jiao, Z. Ma, Z. Ma, C. Chang, X. Yang, R. Wang, *Nanoscale* **2022**, 14, 9681.
- [159] R. Zhao, Y. Ye, Z. Dai, T. Mu, X. Ren, *Results Phys.* **2022**, 43, 106055.
- [160] G. Deng, A. Guo, Z. Kou, J. Yang, Y. Li, Z. Yin, J. Xia, *IEEE Trans. Terahertz Sci. Technol.* **2023**, 14, 57.
- [161] Y. Xie, X. Liu, J. Zhou, H. Zhang, J.-Y. Lin, W. Chen, L.-G. Zhu, K. Meng, Q. H. Liu, J. Zhu, *IEEE Trans. Microw. Theory Tech.* **2023**, 72, 2368.
- [162] Y. Xie, Y. Ma, X. Liu, S. A. Khan, W. Chen, L. Zhu, J. Zhu, Q. H. Liu, *IEEE J. Sel. Top. Quantum Electron.* **2023**, 29, 8600110.
- [163] J. Ajayan, S. Sreejith, M. Manikandan, W.-C. Lai, S. Saha, *Sens. Actuators A: Phys.* **2024**, 369, 115169.
- [164] Y. Cao, H. Sun, Y. Chen, L. Ma, L. Li, S. Jin, W. Wu, *Nanoscale* **2024**, 16, 9055.
- [165] C. Li, X. Xiang, P. Wang, Y. Teng, H. Chen, W. Li, S. Yang, B. Chen, C. Zhang, J. Wu, *Opt. Express* **2024**, 32, 27473.
- [166] S. Ma, Y. Zhang, Z. Li, Z. Wang, L. Liang, X. Yan, H. Yao, X. Hu, G. Wu, F. Huai, *Opt. Express* **2024**, 32, 34764.
- [167] S. Shen, L. Chen, *IEEE Sens. J.* **2024**, 24, 16070.
- [168] W. Xing, X. Bian, R. Su, Y. Han, M. Liang, R. You, *IEEE Sens. J.* **2024**, 24, 34311.
- [169] Y. Xu, X. Chen, S. Zhang, Z. Tian, *Opt. Laser Technol.* **2024**, 177, 110959.
- [170] Y. Yuan, T. Zhang, Z. Zhang, X. Zhao, X. Wu, S. Zheng, L. Liang, C. Cao, *IEEE Sens. J.* **2024**, 24, 36706.
- [171] Z. Zhang, Z. Wang, C. Zhang, Z. Yao, S. Zhang, R. Wang, Z. Tian, J. Han, C. Chang, J. Lou, X. Yan, C. Qiu, *Adv. Mater.* **2024**, 36, 2308453.
- [172] X. Liu, Y. Xie, Y. Yan, Q. Niu, L.-G. Zhu, Z. Dong, Q. H. Liu, J. Zhu, *ACS Photonics* **2024**, 11, 4838.
- [173] C. Yang, Z. Shen, D. Han, Y. Hou, X. Ji, L. Zhang, R. Yan, X. Chen, *Opt. Laser Technol.* **2025**, 182, 112139.
- [174] G. Zhu, F. Huang, Z. Dai, X. Ju, S. Zhong, X. Wang, *Micromachines* **2021**, 12, 1093.
- [175] W. He, M. Tong, Z. Xu, Y. Hu, X. Cheng, T. Jiang, *Photonics Res.* **2021**, 9, 1099.
- [176] X. Ju, G. Zhu, F. Huang, Z. Dai, Y. Chen, C. Guo, L. Deng, X. Wang, *Opt. Express* **2022**, 30, 957.
- [177] B. Hermann, U. Haeusler, G. Yadav, A. Kirchner, T. Feurer, C. Welsch, P. Hommelhoff, R. Ischebeck, *ACS Photonics* **2022**, 9, 1143.
- [178] H.-T. Lee, J. Kim, H.-R. Park, *Opt. Mater. Express* **2023**, 13, 3384.
- [179] H.-T. Lee, J. Kim, J. S. Lee, M. Yoon, H.-R. Park, *Nano Lett.* **2023**, 23, 11685.
- [180] L. Wang, H. Liu, T. Li, D. Tian, C. Zhang, J. Wu, K. Fan, B. Jin, J. Chen, P. Wu, *Appl. Phys. Lett.* **2024**, 124, 131701.
- [181] P. Fu, P. Pai, W. Lan, S. Du, J. Shao, L. Hu, C. Li, Y. Zhang, Z.-L. Deng, B. Liu, L. Yang, C. Gu, *Nano Lett.* **2025**, 25, 3773.
- [182] M. van Exter, C. Fattinger, D. Grischkowsky, *Opt. Lett.* **1989**, 14, 1128.

- [183] G. Ji, H. S. Kim, S. H. Cha, H.-T. Lee, H. J. Kim, S. W. Lee, K. J. Ahn, K.-H. Kim, Y. H. Ahn, H.-R. Park, *Nanophotonics* **2023**, 12, 147.
- [184] W. Shi, C. Li, H. Wang, Z. Wang, L. Yang, *iScience* **2023**, 26, 105871.
- [185] L. Zhao, Y. Cui, J. Li, Y. Xie, W. Li, J. Zhang, *Nanomaterials* **2023**, 13, 1839.
- [186] J. Kim, J. An, W. Kim, J. Seong, Y. Park, E. Lee, S. Kim, S. Moon, C.-K. Lee, H. Lee, J. Rho, *Laser Photonics Rev.* **2025**, 19, 2401425.
- [187] H.-T. Lee, H. Lee, J. Kim, M. Park, C. Sohn, H.-R. Park, *Nanophotonics* **2025**, 14, 1257.
- [188] E. Khaidarov, D. Eschimese, K. H. Lai, A. Huang, Y. H. Fu, Q. Lin, R. Paniagua-Dominguez, A. I. Kuznetsov, *Sci. Rep.* **2022**, 12, 14044.
- [189] Y. Yang, E. Lee, Y. Park, J. Seong, H. Kim, H. Kang, D. Kang, D. Han, J. Rho, *ACS Nano* **2025**, 19, 3008.
- [190] J. Kim, J. Seong, W. Kim, G.-Y. Lee, S. Kim, H. Kim, S.-W. Moon, D. K. Oh, Y. Yang, J. Park, J. Jang, Y. Kim, M. Jeong, C. Park, H. Choi, G. Jeon, K. Lee, D. H. Yoon, N. Park, B. Lee, H. Lee, J. Rho, *Nat. Mater.* **2023**, 22, 474.
- [191] K. Men, Z. Lian, H. Tu, H. Zhao, Q. Wei, Q. Jin, C. Mao, F. Wei, *Micromachines* **2023**, 15, 53.
- [192] S. J. Park, J. T. Hong, S. J. Choi, H. S. Kim, W. K. Park, S. T. Han, J. Y. Park, S. Lee, D. S. Kim, Y. H. Ahn, *Sci. Rep.* **2014**, 4, 4988.
- [193] T. Lang, Z. Yu, J. Zhang, Z. Hong, J. Liu, P. Wang, *Sens. Actuators A: Phys.* **2023**, 360, 114522.
- [194] Y. Yao, Z. Wang, Z. Zhu, Y. He, S. Sun, L. Zhou, Q. He, *Appl. Sci.* **2023**, 13, 9607.
- [195] X. Chen, H.-R. Park, M. Pelton, X. Piao, N. C. Lindquist, H. Im, Y. J. Kim, J. S. Ahn, K. J. Ahn, N. Park, D.-S. Kim, S.-H. Oh, *Nat. Commun.* **2013**, 4, 2361.
- [196] S. Peng, X. Lu, L. Tang, X. Chang, J. Yan, Q. Shi, K. Chen, J. Li, L. Du, W. Huang, *Opt. Express* **2023**, 31, 2644.
- [197] C. Lan, D. Zhu, J. Gao, B. Li, Z. Gao, *Opt. Express* **2018**, 26, 11633.
- [198] J. Gao, C. Lan, Q. Zhao, B. Li, J. Zhou, *Opt. Express* **2018**, 26, 13001.
- [199] P. Zhu, W. Yang, R. Wang, S. Gao, B. Li, Q. Li, *Adv. Opt. Mater.* **2017**, 5, 1600977.
- [200] F. Feng, R. Yan, X. Yang, R. Wang, *Opt. Mater.* **2023**, 141, 113924.
- [201] S. Li, Z. Shen, W. Yin, L. Zhang, X. Chen, *J. Phys. D: Appl. Phys.* **2022**, 55, 325003.
- [202] W. Yin, Z. Shen, Y. Cui, H. Hao, H. Zhang, S. Li, F. Gao, S. Fan, L. Zhang, X. Chen, *Opt. Lett.* **2022**, 47, 5513.
- [203] C. Maibohm, O. F. Silvestre, J. Borme, M. Sinou, K. Heggarty, J. B. Nieder, *Sci. Rep.* **2020**, 10, 8740.
- [204] V. Harinarayana, Y. C. Shin, *Opt. Laser Technol.* **2021**, 142, 107180.



Xiaotong Li is a PhD student in Mechanical Engineering at Pohang University of Science and Technology (POSTECH). He completed his B.Sc. in Electronic Information Engineering in 2019 from Shanghai Dianji University, China, and his M.S. in Communication Engineering in 2023 from Shanghai University, China. His research interests include metasensors, metalens, and metasurface holography.



Hyoung-Taek Lee received his Ph.D. in Physics from UNIST in 2024. He is currently a postdoctoral researcher at the Center for Flat Optics and Metaphotonics, POSTECH. His research focuses on terahertz nanophotonics, active metasurfaces, and inverse design techniques. He is particularly interested in developing tunable and reconfigurable photonic devices for sensing, modulation, and signal processing applications.



Doohyuk Han is a Ph.D. student in Mechanical Engineering at POSTECH. He received a B.S. degree in Physics from Kyungpook National University in 2022 and received his M.S. degree in Physics from UNIST in 2024. His research focuses on nonlinear metamaterials, high-Q resonator, and quantum light generation.



Junsuk Rho is currently a Yeon-San Endowed Chair Professor and Mu-Eun-Jae Endowed Chair Professor at Pohang University of Science and Technology (POSTECH), Korea, with a joint appointment in the Department of Chemical Engineering, the Department of Mechanical Engineering and the Department of Electrical Engineering. He received his Ph.D. at the University of California, Berkeley (2013), M.S. at the University of Illinois, Urbana-Champaign (2008) and B.S. at Seoul National University, Korea (2007) all in Mechanical Engineering.